AIXI

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Prerequisites

- Solomonoff Induction — **start there first.** The Bayesian mixture, posterior updates, and dominance carry over directly; AIXI reuses them in the action setting.
- Comfort with discrete probability (conditional distributions, chain rule, expectations).
- Familiarity with sequential decision-making / reinforcement learning (agents, rewards, discounting, value functions) is useful but not assumed.

What you'll learn

- Formalize the agent-environment interaction, value functions, and the Bayes-optimal policy that defines AIXI.
- Derive the explicit expectimax form of AIXI.
- Prove on-policy value convergence and that AIXI cannot be fooled in deterministic environments.
- Prove the self-optimizing property via likelihood-ratio martingales and change of measure.

Much of this material is drawn from [HQC24]. The book provides fuller explanations, additional context, and proofs omitted here.

Difficulty ratings. Each subproblem is tagged with a rating in square brackets using Knuth's scale [Knu73]: roughly, **[00]** trivial, **[10]** 15-minute pencil-and-paper, **[20]** 1–2 hours, **[30]** several hours to a day, **[40]** a significant research result. Intermediate values are possible. See Appendix B for the full scale.

Problems marked $(*)$ are less interesting/insightful, and while the result may be used later, I would recommend skipping them on a first pass.

Overview

AIXI is the *action* half of **Universal AI**: it lifts the Bayesian mixture from Solomonoff induction — learning to predict — to learning to act. Now the agent takes actions that shape what it observes, and AIXI is the **Bayes-optimal policy** over a universal mixture ξ of environments. It learns to act as well as if it knew the true environment μ. Under the universal choice — $\mathcal{M}$ = all lower-semicomputable environments, prior $w_\nu = 2^{-K(\nu)}$ — the assumption "$\mu \in \mathcal{M}$" again becomes "the universe is computable."

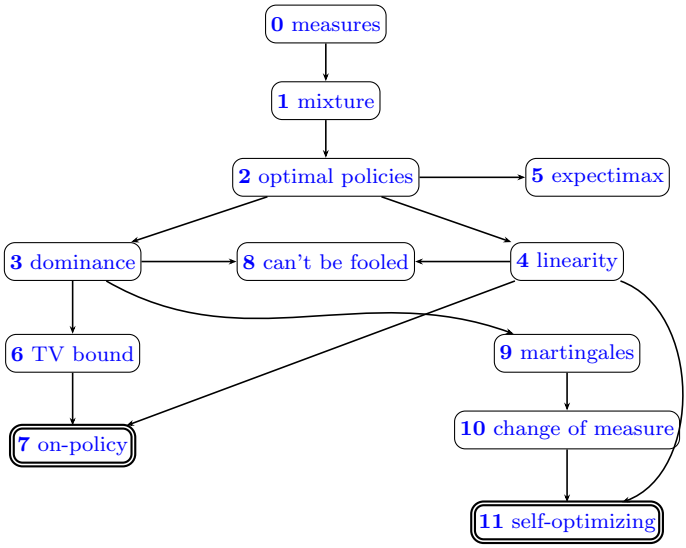
Goal and Roadmap

This exercise sheet builds towards three main results:

- **On-policy value convergence** (Section 7): the Bayesian mixture ξ learns to predict the value of any fixed policy as well as the true environment μ.
- **AIXI can't be fooled** (Section 8): in deterministic environments, the Bayes-optimal agent is guaranteed non-zero value whenever optimal value is non-zero.
- **Self-optimizing property** (Section 11, advanced stretch goal): the Bayes-optimal policy π^*_ξ learns to *act* as well as if it knew μ, provided that any learnable policy can achieve this. This is the central theoretical justification for the AIXI agent.

A good target is to complete Sections 7 and 8. The self-optimizing property (Sections 9–11) requires substantial additional machinery (supermartingales, change of measure) and is an advanced stretch goal.

Critical path. The three results share a common foundation (Sections 0–4) and then diverge:



In summary:

- Sections 0–4 establish the Bayesian RL framework: measure algebra, mixture properties, existence of optimal policies, dominance, and linearity.
- Section 5 derives the explicit expectimax form of AIXI.
- Sections 6–7 prove *on-policy value convergence*: V_ξ^π and V_μ^π become indistinguishable for any fixed π.
- Section 8 shows *AIXI can't be fooled*: the Bayes-optimal agent achieves non-zero value whenever optimal value is non-zero (in deterministic environments).
- Sections 9–11 (advanced stretch goal) prove the *self-optimizing property*: if any policy can learn to act optimally, π_ξ^* inherits this.

Setup

An **agent** interacts with an **environment** in discrete time steps $t = 1, 2, \ldots$.

Definition 0.1 (Spaces and notation)**.**

- $\mathcal{A}$: finite set of **actions**
- $\mathcal{O}$: finite set of **observations**
- $\mathcal{R} \subset [0, 1]$: finite set of **rewards**
- $\mathcal{E} := \mathcal{O} \times \mathcal{R}$: set of **percepts**; $e_t = (o_t, r_t) \equiv o_t r_t$

- $\mathcal{H}^t := (\mathcal{A} \times \mathcal{E})^t$: set of all histories of length t
- $\mathcal{H}^* := \cup_{t=0}^{\infty} \mathcal{H}^t$: set of all finite **histories**
- $\mathcal{H} \equiv \mathcal{H}^*$: set of all finite histories
- $\mathcal{H}^\infty := (\mathcal{A} \times \mathcal{E})^\infty$: set of all infinite histories
- ΔS: set of all probability distributions over set S
- $[\![\cdot]\!]$: **Iverson bracket**: $[\![P]\!] = 1$ if P is true, 0 if false
- t: current time step; m: finite horizon; $1 \leq t \leq m$
- i, j, k: arbitrary integer indices
- xy: concatenation
- ϵ: the empty string/history
- $æ_{i:j} := a_i e_i \, a_{i+1} e_{i+1} \, \cdots \, a_j e_j$: history segment from time i to j
- $æ_{<t} := a_1 e_1 \, a_2 e_2 \, \cdots \, a_{t-1} e_{t-1}$: history up to (but not including) time t

Definition 0.2 (Policy π)**.** A **policy** $\pi : \mathcal{H} \to \Delta\mathcal{A}$ maps each history to a probability distribution over actions. Given history $æ_{<t}$:

- $\pi(\cdot \mid æ_{<t})$ is a distribution over $\mathcal{A}$,
- $\pi(a_t \mid æ_{<t}) \in [0, 1]$ is the probability of choosing action a_t,
- the agent samples $a_t \sim \pi(\cdot \mid æ_{<t})$.

A policy is **deterministic** if $\pi(a \mid æ_{<t}) \in \{0, 1\}$ for all $a, æ_{<t}$. We write $\pi(æ_{i:j}) := \prod_{k=i}^{j} \pi(a_k \mid æ_{<k})$.

Definition 0.3 (Environment ν)**.** An **environment** $\nu : \mathcal{H} \times \mathcal{A} \to \Delta\mathcal{E}$ maps each history–action pair to a distribution over percepts. Given history $æ_{<t}$ and action a_t:

- $\nu(\cdot \mid æ_{<t} a_t)$ is a distribution over $\mathcal{E}$,
- $\nu(e_t \mid æ_{<t} a_t) \in [0, 1]$ is the probability of percept e_t,
- the environment samples $e_t \sim \nu(\cdot \mid æ_{<t} a_t)$.

We write $\nu(æ_{i:j}) := \prod_{k=i}^{j} \nu(e_k \mid æ_{<k} a_k)$. This satisfies the chain rule: $\nu(æ_{i:j}) = \nu(æ_{i:j-1}) \cdot \nu(e_j \mid æ_{i:j-1} a_j)$ or in the form we will usually use, $\nu(æ_{1:t}) = \nu(æ_{<t}) \cdot \nu(e_t \mid æ_{<t} a_t)$. An environment is **deterministic** if $\nu(e_t \mid æ_{<t} a_t) \in \{0, 1\}$ for

all $e_t, æ_{<t}, a_t$ (each percept is produced with certainty). We denote the true (unknown) environment by μ.

Definition 0.4 (Interaction measure ν^π)**.** When policy π interacts with environment ν, the joint probability of a history segment $æ_{i:j}$ given past $æ_{<i}$ is

$$\nu^\pi(æ_{i:j} \mid æ_{<i}) \;:=\; \prod_{k=i}^{j} \pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k} a_k).$$

0. Properties of Measures

Exercise 0.1 (Factorization). **[05]** Show that $\nu^\pi(æ_{1:t}) = \pi(æ_{1:t}) \cdot \nu(æ_{1:t})$.

Solution to Exercise 0.1

Expanding the definition: $\nu^\pi(æ_{1:t}) = \prod_{k=1}^{t} \pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k} a_k) = \underbrace{\prod_{k=1}^{t} \pi(a_k \mid æ_{<k})}_{\pi(æ_{1:t})} \cdot \underbrace{\prod_{k=1}^{t} \nu(e_k \mid æ_{<k} a_k)}_{\nu(æ_{1:t})}$.

Key observation: $\pi(æ_{1:t})$ is the same regardless of the environment. When comparing ν^π and μ^π, the policy factors cancel.

Exercise 0.2 (Chain rule). **[05]** Show that $\nu(æ_{1:t}) = \nu(æ_{<t}) \cdot \nu(e_t \mid æ_{<t} a_t)$.

Solution to Exercise 0.2

From the definition $\nu(æ_{1:t}) = \prod_{k=1}^{t} \nu(e_k \mid æ_{<k} a_k)$, split off the last factor:

$$\nu(æ_{1:t}) \;=\; \prod_{k=1}^{t-1} \nu(e_k \mid æ_{<k} a_k) \cdot \nu(e_t \mid æ_{<t} a_t) \;=\; \nu(æ_{<t}) \cdot \nu(e_t \mid æ_{<t} a_t).$$

Exercise 0.3 (Marginalizing percepts). **[03]** $(*)$ Show that $\sum_{e_t} \nu^\pi(a_t e_t \mid æ_{<t}) = \pi(a_t \mid æ_{<t})$.

Solution to Exercise 0.3

From Definition 0.4, the one-step interaction is $\nu^\pi(a_t e_t \mid æ_{<t}) = \pi(a_t \mid æ_{<t})\, \nu(e_t \mid æ_{<t} a_t)$. Summing over e_t:

$$\sum_{e_t} \nu^\pi(a_t e_t \mid æ_{<t}) \;=\; \pi(a_t \mid æ_{<t}) \underbrace{\sum_{e_t} \nu(e_t \mid æ_{<t} a_t)}_{=1} \;=\; \pi(a_t \mid æ_{<t}).$$

Interpretation: after summing out the environment's response, only the agent's action probability remains.

Exercise 0.4 (Deterministic interaction measure). **[05]** $(*)$ Let π be a deterministic policy, i.e. at each history $æ_{<k}$ there is a unique action a_k^* with $\pi(a_k^* \mid æ_{<k}) = 1$. Show that, provided $æ_{i:j}$ is consistent with π (i.e. $a_k = a_k^*$ for every $k \in \{i, \dots, j\}$):

$$\nu^\pi(æ_{i:j} \mid æ_{<i}) \;=\; \nu(æ_{i:j} \mid æ_{<i}),$$

i.e. on π-consistent futures the interaction measure reduces to the environment measure: every policy factor is 1.

Solution to Exercise 0.4

Since π is deterministic, $\pi(a_k \mid æ_{<k}) = [\![a_k = a_k^*]\!]$. By hypothesis $æ_{i:j}$ is π-consistent, so every such bracket equals 1. From Definition 0.4:

$$\begin{aligned} \nu^\pi(æ_{i:j} \mid æ_{<i}) \;&=\; \prod_{k=i}^{j} \underbrace{\pi(a_k \mid æ_{<k})}_{=1}\, \nu(e_k \mid æ_{<k} a_k) \\ &=\; \prod_{k=i}^{j} \nu(e_k \mid æ_{<k}\, a_k) \;=\; \nu(æ_{i:j} \mid æ_{<i}), \end{aligned}$$

where the last equality is the natural segment extension of $\nu(æ_{1:t}) := \prod_{k=1}^{t} \nu(e_k \mid æ_{<k} a_k)$.

Exercise 0.5. **[05]** (General chain rule for ν^π) Show that for any $p \le q \le r$:

$$\nu^\pi(æ_{p:r} \mid æ_{<p}) \;=\; \nu^\pi(æ_{p:q} \mid æ_{<p}) \cdot \nu^\pi(æ_{q+1:r} \mid æ_{<p}\, æ_{p:q}).$$

Solution to Exercise 0.5

Split the defining product $\nu^\pi(æ_{p:r} \mid æ_{<p}) = \prod_{k=p}^{r} \pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k}a_k)$ at index q:

$$\begin{aligned}\nu^\pi(æ_{p:r} \mid æ_{<p}) &= \prod_{k=p}^{r} \pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k}a_k) \\ &= \underbrace{\prod_{k=p}^{q} \pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k}a_k)}_{\nu^\pi(æ_{p:q} \mid æ_{<p})} \cdot \underbrace{\prod_{k=q+1}^{r} \pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k}a_k)}_{\nu^\pi(æ_{q+1:r} \mid æ_{<p}\, æ_{p:q})},\end{aligned}$$

identifying each block as a conditional via the same expansion: the past for step $k \geq q+1$ is the given history $æ_{<p}$ extended by the first chunk $æ_{p:q}$.
Note: for a deterministic policy, Exercise 0.4 simplifies this further: each factor $\pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k}a_k)$ becomes just $\nu(e_k \mid æ_{<k}\, a_k^*)$.

The Bayesian Mixture and Value Function

Definition 0.1 (Model class, prior, and Bayesian mixture ξ)**.** Let $\mathcal{M} = \{\nu_1, \nu_2, \ldots\}$ be a countable class of environments with prior weights $w_\nu > 0$ satisfying $\sum_{\nu \in \mathcal{M}} w_\nu \leq 1$. We assume $\mu \in \mathcal{M}$.
The **Bayesian mixture** ξ is defined as

$$\xi(æ_{1:t}) := \sum_{\nu \in \mathcal{M}} w_\nu\, \nu(æ_{1:t}) \quad \text{and} \quad \xi(e_t \mid æ_{<t}a_t) := \xi(æ_{1:t})/\xi(æ_{<t}).$$

One can show that the one-step predictive distribution can be written as

$$\xi(e_t \mid æ_{<t}a_t) = \sum_{\nu \in \mathcal{M}} w(\nu \mid æ_{<t})\, \nu(e_t \mid æ_{<t}a_t),$$

where the **posterior weight** is

$$w(\nu \mid æ_{<t}) := w_\nu \frac{\nu(æ_{<t})}{\xi(æ_{<t})}, \qquad w(\nu \mid \epsilon) := w_\nu.$$

See Appendix A for a worked example.

Definition 0.2 (Expectation $\mathbb{E}^\pi_\nu$)**.** For a function $f : \mathcal{H} \to \mathbb{R}$ of a finite future segment $æ'_{t:m}$:

$$\mathbb{E}^\pi_\nu[f \mid æ_{<t}] \;:=\; \mathop{\mathbb{E}}_{æ_{t:m}\sim\nu^\pi(\cdot|æ_{<t})}[f] \;=\; \sum_{æ'_{t:m}} \nu^\pi(æ'_{t:m} \mid æ_{<t})\, f(æ'_{t:m}).$$

For functions f of the infinite future (like the value function), with a sequence $f_1, f_2, \ldots \to f$, where f_m depends only on $æ'_{t:m}$, we define $\mathbb{E}^\pi_\nu[f \mid æ_{<t}] := \lim_{m\to\infty} \mathbb{E}^\pi_\nu[f_m \mid æ_{<t}]$.

Definition 0.3 (Discounted return $G_{t:m}$)**.** Fix a discount factor $\gamma \in (0, 1)$, held constant throughout. The **discounted return** at time step t, up to horizon m, is

$$G^\gamma_{t:m} \;:=\; \sum_{k=t+1}^{m} \gamma^{k-(t+1)}\, r_k \;=\; r_{t+1} + \gamma\, r_{t+2} + \cdots + \gamma^{m-t}\, r_m.$$

Since γ is fixed we drop it and write $G_{t:m} \equiv G^\gamma_{t:m}$. It satisfies the recursion $G_{t:m} = r_{t+1} + \gamma\, G_{t+1:m}$, with $G_{m-1:m} = r_m$ and $G_{n:m} = 0$ for $n \geq m$. The **infinite-horizon return** is the pointwise limit

$$G_{\geq t} \;\equiv\; G_{t:\infty} \;:=\; \lim_{m\to\infty} G_{t:m} \;=\; \sum_{k=t+1}^{\infty} \gamma^{k-(t+1)}\, r_k,$$

with the clean recursion $G_{\geq t} = r_{t+1} + \gamma\, G_{\geq t+1}$ and no boundary cases.

Definition 0.4 (Value function[a])**.** The **value** of policy π in environment ν with horizon $m \geq t$ given history $æ_{<t}$ is

$$\begin{aligned} V^{\pi,m}_\nu(æ_{<t}) \;&:=\; (1-\gamma)\, \mathbb{E}^\pi_\nu[G_{t-1:m} \mid æ_{<t}] \\ &=\; (1-\gamma) \sum_{æ_{t:m}} \nu^\pi(æ_{t:m} \mid æ_{<t})\, G_{t-1:m} \\ &=\; (1-\gamma) \sum_{æ_{t:m}} \nu^\pi(æ_{t:m} \mid æ_{<t}) \left[\sum_{k=t}^{m} \gamma^{k-t} r_k\right]. \end{aligned}$$

The **infinite-horizon value** is the pointwise limit $V^{\pi,\infty}_\nu(æ_{<t}) := \lim_{m\to\infty} V^{\pi,m}_\nu(æ_{<t}) = (1-\gamma)\, \mathbb{E}^\pi_\nu[G_{\geq t-1} \mid æ_{<t}]$. We write $V^\pi_\nu \equiv V^{\pi,\infty}_\nu$ for short. The **optimal value** is $V^{*,m}_\nu(æ_{<t}) := \sup_\pi V^{\pi,m}_\nu(æ_{<t})$ for $m \in \mathbb{N} \cup \{\infty\}$, with $V^*_\nu \equiv V^{*,\infty}_\nu$.[b] An **optimal policy** π^*_ν satisfies $V^{\pi^*_\nu}_\nu = V^*_\nu$. The **Bayes-optimal policy** is $\pi^*_\xi \in \arg\max_\pi V^\pi_\xi$.

[a] See Appendix A for a worked example. For simplicity, we consider only geometric discounting.

[b] The optimal value is defined as a sup over policies. In general, a supremum need not be attained (e.g. $\sup_{x\in(0,1)} x = 1$ but no $x \in (0,1)$ achieves it). Section 2 shows that the sup is attained in our setting.

Remark (AIXI)

All results in this sheet hold for any countable $\mathcal{M}$ and prior weights w_ν. **AIXI** is a special case of the Bayes-optimal policy π^*_ξ for the particular choice $\mathcal{M} :=$ the class of all lower-semicomputable chronological semimeasures, and prior $w_\nu := 2^{-K(\nu)}$, where $K(\nu)$ is the Kolmogorov complexity of ν: the length of the shortest program that computes ν. The resulting agent is written $\pi^*_{\xi_U}$, or simply AIξ.
Why this $\mathcal{M}$? By including every computable environment, the assumption $\mu \in \mathcal{M}$ reduces to "the universe is computable": as weak an assumption as one can make.
Why this prior? The prior $2^{-K(\nu)}$ is *dominant*: for any other computable prior w'_ν, there exists a constant $c > 0$ such that $2^{-K(\nu)} \geq c \cdot w'_\nu$ for all ν.
Caveat: The constant c depends on the choice of universal Turing machine U, and adversarial choices of U can make AIXI behave arbitrarily badly [LH15].
See [HQC24]: Chapter 2.7 for Kolmogorov complexity, Chapters 3.7–3.8 for the model class and universal prior, and Chapter 7.4 for AIXI itself.

1. Properties of the Bayesian Mixture

Exercise 1.1 (Posterior update). [10] $(*)$ Show that the posterior updates multiplicatively:

$$w(\nu \mid æ_{1:t}) \;=\; w(\nu \mid æ_{<t}) \frac{\nu(e_t \mid æ_{<t}a_t)}{\xi(e_t \mid æ_{<t}a_t)}.$$

Hint: Use the definitions of $w(\nu \mid æ_{1:t})$ and $w(\nu \mid æ_{<t})$, and apply Exercise 0.2.

Solution to Exercise 1.1

From the definition: $w(\nu \mid æ_{1:t}) = w_\nu \cdot \nu(æ_{1:t})/\xi(æ_{1:t})$ and $w(\nu \mid æ_{<t}) = w_\nu \cdot \nu(æ_{<t})/\xi(æ_{<t})$.
Dividing:

$$\frac{w(\nu \mid æ_{1:t})}{w(\nu \mid æ_{<t})} = \frac{\nu(æ_{1:t})}{\nu(æ_{<t})} \cdot \frac{\xi(æ_{<t})}{\xi(æ_{1:t})} = \frac{\nu(e_t \mid æ_{<t}a_t)}{\xi(e_t \mid æ_{<t}a_t)},$$

where we used the chain rule (Exercise 0.2) for both ν and ξ.

Exercise 1.2. **[15]** (One-step predictive distribution of ξ) Starting from $\xi(æ_{1:t}) = \sum_{\nu\in\mathcal{M}} w_\nu \nu(æ_{1:t})$, derive the one-step predictive form:

$$\xi(e_t \mid æ_{<t}a_t) \;=\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\, \nu(e_t \mid æ_{<t}a_t).$$

Hint: Write $\xi(e_t \mid æ_{<t}a_t) = \xi(æ_{1:t})/\xi(æ_{<t})$, expand the numerator, and use Exercise 0.2.

Solution to Exercise 1.2

The one-step conditional is the ratio of joint to marginal: $\xi(e_t \mid æ_{<t}a_t) := \xi(æ_{1:t})/\xi(æ_{<t})$.
Expanding the numerator using $\xi(æ_{1:t}) = \sum_\nu w_\nu \nu(æ_{1:t})$:

$$\begin{aligned}
\xi(e_t \mid æ_{<t}a_t) \;&=\; \frac{\sum_{\nu\in\mathcal{M}} w_\nu\, \nu(æ_{1:t})}{\xi(æ_{<t})} \\
&=\; \frac{\sum_\nu w_\nu\, \nu(æ_{<t}) \cdot \nu(e_t \mid æ_{<t}a_t)}{\xi(æ_{<t})} \qquad \text{(chain rule, Exercise 0.2)} \\
&=\; \sum_\nu \underbrace{\frac{w_\nu\, \nu(æ_{<t})}{\xi(æ_{<t})}}_{=\, w(\nu \mid æ_{<t})} \cdot \nu(e_t \mid æ_{<t}a_t) \\
&=\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\, \nu(e_t \mid æ_{<t}a_t).
\end{aligned}$$

Exercise 1.3 (Bounded value). **[10]** Show that $V_\nu^{\pi,m}(æ_{<t}) \in [0,1]$ for any $\nu, \pi, m \in \mathbb{N}\cup\{\infty\}, æ_{<t}$.

Hint: Recall the formula for geometric series: $(1-\gamma)\sum_{k=0}^{n}\gamma^k = 1-\gamma^{n+1}$.

Solution to Exercise 1.3

Since $r_k \in [0,1]$, each discounted reward sum is bounded:

$$0 \;\le\; (1-\gamma)\, G_{t-1:m} \;\le\; (1-\gamma)\sum_{k=t}^{m}\gamma^{k-t} \;=\; 1-\gamma^{m-t+1} \;\le\; 1.$$

The interaction measure satisfies $\nu^\pi(æ'_{t:m} \mid æ_{<t}) \geq 0$ and $\sum_{æ'_{t:m}} \nu^\pi(æ'_{t:m} \mid æ_{<t}) = 1$, so the value function is a weighted average of terms in $[0, 1]$:

$$0 \;\leq\; V_\nu^{\pi,m}(æ_{<t}) \;=\; \sum_{æ'_{t:m}} \nu^\pi(æ'_{t:m} \mid æ_{<t})\,(1-\gamma)\,G_{t-1:m} \;\leq\; 1.$$

For $m = \infty$: since $0 \leq V_\nu^{\pi,m}(æ_{<t}) \leq 1$ for all finite m, the limit $V_\nu^\pi(æ_{<t}) = \lim_{m\to\infty} V_\nu^{\pi,m}(æ_{<t})$ also lies in $[0, 1]$.

Exercise 1.4. **[15]** **(∗)** (Multi-step posterior linearity of ξ^π) Extend Exercise 1.2 to multi-step histories: show that for finite $m \geq t$,

$$\xi^\pi(æ_{t:m} \mid æ_{<t}) \;=\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\,\nu^\pi(æ_{t:m} \mid æ_{<t}).$$

Hint: Apply the general chain rule (Exercise 0.5) to $\xi(æ_{1:m}) = \sum_\nu w_\nu \nu(æ_{1:m})$, then use factorization (Exercise 0.1).

Solution to Exercise 1.4

Start from the definition $\xi(æ_{1:m}) = \sum_{\nu\in\mathcal{M}} w_\nu\,\nu(æ_{1:m})$. Apply the general chain rule (Exercise 0.5) to both sides: $\xi(æ_{1:m}) = \xi(æ_{<t}) \cdot \xi(æ_{t:m} \mid æ_{<t})$ and $\nu(æ_{1:m}) = \nu(æ_{<t}) \cdot \nu(æ_{t:m} \mid æ_{<t})$. Dividing by $\xi(æ_{<t})$:

$$\begin{aligned}\xi(æ_{t:m} \mid æ_{<t}) \;&=\; \sum_{\nu\in\mathcal{M}} \frac{w_\nu\,\nu(æ_{<t})}{\xi(æ_{<t})}\nu(æ_{t:m} \mid æ_{<t}) \\ &=\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\,\nu(æ_{t:m} \mid æ_{<t}).\end{aligned}$$

Now multiply both sides by $\pi(æ_{t:m} \mid æ_{<t})$. By factorization (Exercise 0.1), $\pi(æ_{t:m} \mid æ_{<t}) \cdot \xi(æ_{t:m} \mid æ_{<t}) = \xi^\pi(æ_{t:m} \mid æ_{<t})$ and likewise for each ν:

$$\xi^\pi(æ_{t:m} \mid æ_{<t}) \;=\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\,\nu^\pi(æ_{t:m} \mid æ_{<t}).$$

2. Existence of Optimal Policies

Recall that $V_\nu^*(æ_{<t}) := \sup_\pi V_\nu^\pi(æ_{<t})$ is defined as a supremum over all policies. In general, a supremum need not be achieved: for example, $\sup_{x\in(0,1)} x = 1$, but no $x \in (0, 1)$ attains this value. This problem shows that in our setting, the supremum *is*

attained, so an optimal policy π^*_ν with $V_\nu^{\pi^*_\nu} = V^*_\nu$ exists.

Exercise 2.1. **[05]** Consider a single time step. Given history $æ_{<t}$ and a function $Q : \mathcal{A} \to \mathbb{R}$, show that $\sup_\pi \sum_{a\in\mathcal{A}} \pi(a \mid æ_{<t})\, Q(a) = \max_{a\in\mathcal{A}} Q(a)$, and that the supremum is attained by the deterministic policy that places all probability on an action achieving the maximum.

Solution to Exercise 2.1

Let $a^* \in \arg\max_{a'\in\mathcal{A}} Q(a')$, which exists because $\mathcal{A}$ is finite. So $Q(a^*) = \max_{a'} Q(a')$.
Upper bound. For any π: $\sum_a \pi(a \mid æ_{<t})Q(a) \le \sum_a \pi(a \mid æ_{<t})Q(a^*) = Q(a^*)$.
Lower bound. The deterministic π^* with $\pi^*(a^* \mid æ_{<t}) = 1$ achieves $Q(a^*)$.
Combining: $\sup_\pi \sum_a \pi(a)Q(a) = Q(a^*) = \max_{a'} Q(a')$, attained by π^*.

Exercise 2.2 (Bellman equation). **[15]** Show that for finite m and $t \le m$:

$$V_\nu^{\pi,m}(æ_{<t}) \;=\; \sum_{æ'_t} \nu^\pi(æ'_t \mid æ_{<t}) \Big[(1-\gamma)\, r'_t \;+\; \gamma\, V_\nu^{\pi,m}(æ_{<t}\, æ'_t)\Big],$$

Hint: Use the general chain rule (Exercise 0.5) to peel off the *first* step,

$$\nu^\pi(æ_{t:m} \mid æ_{<t}) \;=\; \nu^\pi(æ_t \mid æ_{<t}) \cdot \nu^\pi(æ_{t+1:m} \mid æ_{1:t}).$$

Break up the sum $\sum_{æ_{t:m}} = \sum_{æ_t} \sum_{æ_{t+1:m}}$, and factor out r'_t using $\sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ) = 1$ for any history $æ$.

Solution to Exercise 2.2

Step 1: Factor the future. Write $æ'_{t:m} = æ'_t\, æ'_{t+1:m}$: $\nu^\pi(æ'_{t:m} \mid æ_{<t}) = \nu^\pi(æ'_t \mid æ_{<t}) \cdot \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t)$.
Step 2: Split the reward sum. This is just the return recursion (Definition 0.3): $G_{t-1:m} = r'_t + \gamma\, G_{t:m}$, where $G_{t-1:m} = \sum_{k=t}^m \gamma^{k-t} r'_k$ is the return given $æ_{<t}$.
Step 3: Substitute into the definition of $V_\nu^{\pi,m}(æ_{<t})$.

$$\begin{aligned} V_\nu^{\pi,m}(æ_{<t}) \;=\; (1-\gamma) \sum_{æ'_t} \nu^\pi(æ'_t \mid æ_{<t}) \sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t) \\ \times \left[r'_t + \gamma G_{t:m}\right]. \end{aligned}$$

Consider the inner term $\sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t)\,[r'_t + \gamma G_{t:m}]$. We can expand as:

$$
\begin{aligned}
&= \sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t) r'_t + \gamma \sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t) G_{t:m} \\
&= r'_t \cancel{\sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t)} + \gamma \sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t) G_{t:m} \\
&= r'_t + \gamma \sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t) G_{t:m}
\end{aligned}
$$

as r'_t doesn't depend on $æ'_{t+1:m}$, and $\sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t) = 1$. Inserting this above, we obtain:

$$
\begin{aligned}
V_\nu^{\pi,m}(æ_{<t}) = \sum_{æ'_t} \nu^\pi(æ'_t \mid æ_{<t}) \Bigg[&(1-\gamma)\, r'_t \\
&+ \gamma \underbrace{(1-\gamma) \sum_{æ'_{t+1:m}} \nu^\pi(æ'_{t+1:m} \mid æ_{<t}\, æ'_t) G_{t:m}}_{= V_\nu^{\pi,m}(æ_{<t}\, æ'_t)} \Bigg].
\end{aligned}
$$

Exercise 2.3 (Backward induction). [20] (∗) Using the Bellman equation from Exercise 2.2 and Exercise 2.1, show by backward induction on $t = m, m-1, \ldots, 1$ that for each finite m, a deterministic optimal policy exists.

Solution to Exercise 2.3

We induct on $t = m, m-1, \ldots, 1$, showing that for every history $æ_{<t}$, a deterministic policy achieves $V_\nu^{*,m}(æ_{<t})$.

Base case ($t = m$): At $t = m$ the continuation term vanishes (the value from time $m+1$ is an empty sum), so the Bellman equation gives $V_\nu^{\pi,m}(æ_{<m}) = (1-\gamma)\sum_{æ'_m} \nu^\pi(æ'_m \mid æ_{<m})\, r'_m$. Splitting the step $æ'_m = a'_m e'_m$ via $\nu^\pi(æ'_m \mid æ_{<m}) = \pi(a'_m \mid æ_{<m})\, \nu(e'_m \mid æ_{<m} a'_m)$ gives $V_\nu^{\pi,m}(æ_{<m}) = (1-\gamma)\sum_{a'_m} \pi(a'_m \mid æ_{<m}) \sum_{e'_m} \nu(e'_m \mid æ_{<m} a'_m)\, r'_m$. This has the form $\sum_a \pi(a) Q(a)$ with $Q(a) = (1-\gamma)\sum_{e'_m} \nu(e'_m \mid æ_{<m} a)\, r'_m$, which is independent of π. By Exercise 2.1, the supremum over π is $\max_a Q(a)$, attained by the deterministic policy playing $a^* \in \arg\max_a Q(a)$.

Inductive step: Suppose that for every history of length t, there exists a deterministic policy π^*_{t+1} achieving $V_\nu^{\pi^*_{t+1},m} = V_\nu^{*,m}$ from time $t+1$ onwards. Substituting

this optimal continuation into the Bellman equation from Exercise 2.2 gives

$$V_\nu^{\pi,m}(æ_{<t}) \;=\; \sum_{æ'_t} \nu^\pi(æ'_t \mid æ_{<t}) \big[(1-\gamma)\, r'_t + \gamma\, V_\nu^{*,m}(æ_{<t}\, æ'_t)\big].$$

Splitting the step $æ'_t = a'_t e'_t$ via $\nu^\pi(æ'_t \mid æ_{<t}) = \pi(a'_t \mid æ_{<t})\, \nu(e'_t \mid æ_{<t} a'_t)$ and grouping the percept sum into the action weight:

$$V_\nu^{\pi,m}(æ_{<t}) \;=\; \sum_{a'_t} \pi(a'_t \mid æ_{<t})\, Q(a'_t),$$

where $Q(a'_t) := \sum_{e'_t} \nu(e'_t \mid æ_{<t} a'_t)[(1-\gamma)\, r'_t + \gamma\, V_\nu^{*,m}(æ_{<t}\, a'_t\, e'_t)]$ is independent of π (the continuation value $V_\nu^{*,m}$ is fixed by the inductive hypothesis). By Exercise 2.1, $\sup_\pi \sum_{a'_t} \pi(a'_t)\, Q(a'_t) = \max_{a'_t} Q(a'_t)$, attained by the deterministic policy playing $a^*_t \in \arg\max_{a'_t} Q(a'_t)$ at time t, then following π^*_{t+1}.

Exercise 2.4 (Bellman optimality equation). **[10]** Using Exercise 2.3, show that for finite m:

$$V_\nu^{*,m}(æ_{<t}) \;=\; \max_{a'_t} \sum_{e'_t} \nu(e'_t \mid æ_{<t} a'_t) \Big[(1-\gamma)\, r'_t \;+\; \gamma\, V_\nu^{*,m}(æ_{<t}\, æ'_t)\Big].$$

where $e'_t = o'_t r'_t$ and $æ'_t = a'_t e'_t$.

Solution to Exercise 2.4

By Exercise 2.3, a deterministic optimal policy π^*_ν exists with $V_\nu^{\pi^*_\nu,m} = V_\nu^{*,m}$. Substituting $\pi = \pi^*_\nu$ into the Bellman equation from Exercise 2.2 (using $V_\nu^{\pi^*_\nu,m} = V_\nu^{*,m}$ on both sides):

$$V_\nu^{*,m}(æ_{<t}) \;=\; \sum_{æ'_t} \nu^{\pi^*_\nu}(æ'_t \mid æ_{<t}) \Big[(1-\gamma)\, r'_t + \gamma\, V_\nu^{*,m}(æ_{<t}\, æ'_t)\Big].$$

Splitting the step $æ'_t = a'_t e'_t$ via $\nu^{\pi^*_\nu}(æ'_t \mid æ_{<t}) = \pi^*_\nu(a'_t \mid æ_{<t})\, \nu(e'_t \mid æ_{<t} a'_t)$:

$$\begin{aligned} V_\nu^{*,m}(æ_{<t}) \;=\; & \sum_{a'_t} \pi^*_\nu(a'_t \mid æ_{<t}) \\ & \times \sum_{e'_t} \nu(e'_t \mid æ_{<t} a'_t) \Big[(1-\gamma)\, r'_t + \gamma\, V_\nu^{*,m}(æ_{<t}\, a'_t\, e'_t)\Big]. \end{aligned}$$

Since π^*_ν is deterministic, let a^*_t denote the unique action with $\pi^*_\nu(a^*_t \mid æ_{<t}) = 1$. Then:

$$V^{*,m}_\nu(æ_{<t}) \;=\; \sum_{e'_t} \nu(e'_t \mid æ_{<t}a^*_t)\Big[(1-\gamma)\, r'_t + \gamma\, V^{*,m}_\nu(æ_{<t}\, a^*_t\, e'_t)\Big].$$

Define $Q(a) := \sum_{e'_t} \nu(e'_t \mid æ_{<t}a)[(1-\gamma)\, r'_t + \gamma\, V^{*,m}_\nu(æ_{<t}\, a\, e'_t)]$. Then $V^{*,m}_\nu(æ_{<t}) = Q(a^*_t)$. We must have $Q(a^*_t) = \max_a Q(a)$: if some $\tilde{a}$ had $Q(\tilde{a}) > Q(a^*_t)$, then the policy that plays $\tilde{a}$ at $æ_{<t}$ and follows π^*_ν elsewhere would achieve value $Q(\tilde{a}) > V^{*,m}_\nu(æ_{<t})$, contradicting $V^{*,m}_\nu = \sup_\pi V^{\pi,m}_\nu$. Hence:

$$V^{*,m}_\nu(æ_{<t}) \;=\; \max_{a'_t} \sum_{e'_t} \nu(e'_t \mid æ_{<t}a'_t)\Big[(1-\gamma)\, r'_t + \gamma\, V^{*,m}_\nu(æ_{<t}\, a'_t\, e'_t)\Big].$$

Exercise 2.5. **[15]** $(*)$ (Existence of V^*_ν) Show that the pointwise limit $V^*_\nu(æ_{<t}) := \lim_{m\to\infty} V^{*,m}_\nu(æ_{<t})$ exists for all histories $æ_{<t}$.

Hint: Show $V^{*,m+1}_\nu(æ_{<t}) \geq V^{*,m}_\nu(æ_{<t})$ by expanding $V^{*,m+1}_\nu$. Use this and Exercise 1.3 and Monotone Convergence Theorem to obtain the result.

Solution to Exercise 2.5

We show $V^{*,m+1}_\nu(æ_{<t}) \geq V^{*,m}_\nu(æ_{<t})$ for all $æ_{<t}$. Since $V^{*,m+1}_\nu = \sup_\pi V^{\pi,m+1}_\nu$, it suffices to show $V^{\pi,m+1}_\nu(æ_{<t}) \geq V^{\pi,m}_\nu(æ_{<t})$ for every π. Expanding the definition:

$$\begin{aligned}
&V^{\pi,m+1}_\nu(æ_{<t}) \\
&= (1-\gamma) \sum_{æ_{t:m+1}} \nu^\pi(æ_{t:m+1} \mid æ_{<t})\, G_{t-1:m+1} \\
&= (1-\gamma) \sum_{æ_{t:m+1}} \nu^\pi(æ_{t:m+1} \mid æ_{<t}) \left[G_{t-1:m} \;+\; \gamma^{m+1-t} r_{m+1}\right] \\
&= (1-\gamma) \sum_{æ_{t:m+1}} \nu^\pi(æ_{t:m+1} \mid æ_{<t}) G_{t-1:m} \\
&\quad + \underbrace{(1-\gamma)\, \gamma^{m+1-t} \sum_{æ_{t:m+1}} \nu^\pi(æ_{t:m+1} \mid æ_{<t})\, r_{m+1}}_{\geq 0}.
\end{aligned}$$

For the first term, marginalize over a_{m+1}, e_{m+1} (which $G_{t-1:m}$ does not depend on):

$$\sum_{æ_{t:m+1}} \nu^\pi(æ_{t:m+1} \mid æ_{<t}) G_{t-1:m} \;=\; \sum_{æ_{t:m}} \nu^\pi(æ_{t:m} \mid æ_{<t}) G_{t-1:m}.$$

So $V_\nu^{\pi,m+1}(æ_{<t}) \geq (1-\gamma)\sum_{æ_{t:m}} \nu^\pi(æ_{t:m} \mid æ_{<t}) G_{t-1:m} = V_\nu^{\pi,m}(æ_{<t})$.
Since $V_\nu^{*,m}(æ_{<t})$ is non-decreasing in m and bounded in $[0,1]$ by Exercise 1.3, the monotone convergence theorem gives that $V_\nu^*(æ_{<t}) := \lim_{m\to\infty} V_\nu^{*,m}(æ_{<t})$ exists for each $æ_{<t}$.

Exercise 2.6 (Infinite-horizon Bellman optimality equation). **[10]** Show that the Bellman optimality equation (Exercise 2.4) extends to $m = \infty$:

$$V_\nu^*(æ_{<t}) \;=\; \max_{a'_t} \sum_{e'_t} \nu(e'_t \mid æ_{<t} a'_t) \Big[(1-\gamma)\, r'_t \;+\; \gamma\, V_\nu^*(æ_{<t}\, a'_t\, e'_t) \Big].$$

Solution to Exercise 2.6

By Exercise 2.4, for every finite m:

$$V_\nu^{*,m}(æ_{<t}) \;=\; \max_{a'_t} \sum_{e'_t} \nu(e'_t \mid æ_{<t} a'_t) \Big[(1-\gamma)\, r'_t + \gamma\, V_\nu^{*,m}(æ_{<t}\, a'_t\, e'_t) \Big].$$

By Exercise 2.5, each $V_\nu^{*,m}(\cdot)$ converges pointwise to $V_\nu^*(\cdot)$ as $m \to \infty$. Since $\mathcal{A}$ and $\mathcal{E}$ are finite, the max and $\sum$ are over finitely many convergent terms, so:

$$V_\nu^*(æ_{<t}) \;=\; \max_{a'_t} \sum_{e'_t} \nu(e'_t \mid æ_{<t} a'_t) \Big[(1-\gamma)\, r'_t + \gamma\, V_\nu^*(æ_{<t}\, a'_t\, e'_t) \Big].$$

Exercise 2.7 (Optimal policy for infinite horizon). **[25]** $(*)$ Show that a deterministic policy π_ν^* exists achieving $V_\nu^{\pi_\nu^*}(æ_{<t}) = V_\nu^*(æ_{<t})$ for all $æ_{<t}$.
Approach:

(a) Define π_ν^* as the greedy policy (the arg max action from Exercise 2.6).

(b) Write two Bellman equations: one for V_ν^*, one for $V_\nu^{\pi_\nu^*,m}$.

(c) Define $\Delta V_\nu^m := V_\nu^* - V_\nu^{\pi_\nu^*,m}$ and obtain a recursive equation for it.

(d) Iterate out to the finite horizon, noting $V_\nu^{\pi_\nu^*,m}(æ_{1:m}) = 0$ (why?)

(e) Bound the tail, take $m \to \infty$.

Solution to Exercise 2.7

Step 1: Define the greedy policy. For each history $æ_{<t}$, let

$$a_t^* \in \arg\max_{a'_t} \sum_{e'_t} \nu(e'_t \mid æ_{<t} a'_t)[(1-\gamma)\, r'_t + \gamma\, V_\nu^*(æ_{<t}\, a'_t\, e'_t)],$$

which exists because $\mathcal{A}$ is finite. Define the deterministic policy $\pi_\nu^*(a \mid æ_{<t}) := [\![a = a_t^*]\!]$. By Exercise 2.6, the infinite-horizon Bellman optimality equation becomes:

$$V_\nu^*(æ_{<t}) \;=\; \sum_{e'_t} \nu(e'_t \mid æ_{<t}\, a_t^*) \Big[(1-\gamma)\, r'_t + \gamma\, V_\nu^*(æ_{<t}\, a_t^*\, e'_t)\Big].$$

Step 2: Error recursion. The Bellman equation (Exercise 2.2) holds for any policy, so for π_ν^* at finite horizon m:

$$V_\nu^{\pi_\nu^*,m}(æ_{<t}) \;=\; \sum_{e'_t} \nu(e'_t \mid æ_{<t}\, a_t^*) \Big[(1-\gamma)\, r'_t + \gamma\, V_\nu^{\pi_\nu^*,m}(æ_{<t}\, a_t^*\, e'_t)\Big].$$

Subtracting this from the equation in Step 1, the $(1-\gamma)\, r'_t$ terms cancel:

$$\begin{aligned} V_\nu^*(æ_{<t}) - V_\nu^{\pi_\nu^*,m}(æ_{<t}) \;&=\; \gamma \sum_{e'_t} \nu(e'_t \mid æ_{<t}\, a_t^*) \\ &\quad \times \Big[V_\nu^*(æ_{<t}\, a_t^*\, e'_t) - V_\nu^{\pi_\nu^*,m}(æ_{<t}\, a_t^*\, e'_t)\Big]. \end{aligned}$$

The error at time t is γ times a weighted average of errors at time $t+1$.
Step 3: Iterate the contraction. Write $\Delta V_\nu^m(æ_{<t}) := V_\nu^*(æ_{<t}) - V_\nu^{\pi_\nu^*,m}(æ_{<t}) \geq 0$ for the error. Step 2 gives:

$$\Delta V_\nu^m(æ_{<t}) \;=\; \gamma \sum_{e'_t} \nu(e'_t \mid æ_{<t}\, a_t^*)\, \Delta V_\nu^m(æ_{<t}\, a_t^*\, e'_t).$$

Applying the same recursion at time $t+1$ (with greedy action a_{t+1}^* at history $æ_{<t}\, a_t^*\, e'_t$):

$$\begin{aligned} \Delta V_\nu^m(æ_{<t}) \;&=\; \gamma^2 \sum_{e'_t} \nu(e'_t \mid æ_{<t}\, a_t^*) \sum_{e'_{t+1}} \nu(e'_{t+1} \mid æ_{<t}\, a_t^*\, e'_t\, a_{t+1}^*) \\ &\quad \times \Delta V_\nu^m(æ_{<t}\, a_t^*\, e'_t\, a_{t+1}^*\, e'_{t+1}). \end{aligned}$$

After $m-t+1$ iterations, summing over all percept sequences $e'_{t:m}$:

$$\Delta V_\nu^m(æ_{<t}) \;=\; \gamma^{m-t+1} \sum_{e'_{t:m}} \left(\prod_{k=t}^{m} \nu(e'_k \mid æ'_{<k}\, a_k^*) \right) \Delta V_\nu^m(æ_{<t}\, æ'_{t:m}),$$

where $æ'_{t:m} = a^*_t\, e'_t \cdots a^*_m\, e'_m$ is the history segment generated by π^*_ν. By Exercise 0.4, the product of ν-conditionals is $\nu^{\pi^*_\nu}(æ'_{t:m} \mid æ_{<t})$. At the boundary, $V_\nu^{\pi^*_\nu,m}(æ_{<t}\, æ'_{t:m}) = 0$ (summing over no rewards past horizon m), so $\Delta V_\nu^m(æ_{<t}\, æ'_{t:m}) = V_\nu^*(æ_{<t}\, æ'_{t:m}) \in [0,1]$. Therefore:

$$\Delta V_\nu^m(æ_{<t}) \;=\; \gamma^{m-t+1} \sum_{æ'_{t:m}} \nu^{\pi^*_\nu}(æ'_{t:m} \mid æ_{<t})\, V_\nu^*(æ_{<t}\, æ'_{t:m}) \;\le\; \gamma^{m-t+1}.$$

Step 4: Take $m \to \infty$. Since $0 \le V_\nu^*(æ_{<t}) - V_\nu^{\pi^*_\nu,m}(æ_{<t}) \le \gamma^{m-t+1} \to 0$, we have $V_\nu^{\pi^*_\nu,m}(æ_{<t}) \to V_\nu^*(æ_{<t})$, i.e. $V_\nu^{\pi^*_\nu} = V_\nu^*$.

3. Dominance of the Bayesian Mixture

Exercise 3.1. **[03]** Show that $\xi(æ_{<t}) \ge w_\nu \cdot \nu(æ_{<t})$ for every $\nu \in \mathcal{M}$.

Solution to Exercise 3.1

$\xi(æ_{<t}) = \sum_{\nu' \in \mathcal{M}} w_{\nu'} \nu'(æ_{<t}) \ge w_\nu \nu(æ_{<t})$.

Exercise 3.2. **[10]** Conclude that $\xi^\pi(æ_{1:m}) \ge w_\nu \cdot \nu^\pi(æ_{1:m})$ for any $\nu \in \mathcal{M}$ and any policy π.

Solution to Exercise 3.2

By Exercise 0.1: $\xi^\pi(æ_{1:m}) = \pi(æ_{1:m}) \cdot \xi(æ_{1:m}) \ge \pi(æ_{1:m}) \cdot w_\nu \nu(æ_{1:m}) = w_\nu \cdot \nu^\pi(æ_{1:m})$.

4. Properties of V_ξ

Exercise 4.1. **[15]** Show that for finite m:

$$V_\xi^{\pi,m}(æ_{<t}) \;=\; \sum_{\nu \in \mathcal{M}} w(\nu \mid æ_{<t})\, V_\nu^{\pi,m}(æ_{<t}).$$

Hint: Use the multi-step posterior linearity of ξ^π (Exercise 1.4).

Solution to Exercise 4.1

By Exercise 1.4: $\xi^\pi(æ_{t:m} \mid æ_{<t}) = \sum_\nu w(\nu \mid æ_{<t})\, \nu^\pi(æ_{t:m} \mid æ_{<t})$. Multiply both sides by $(1-\gamma)G_{t-1:m}$ and sum over all $æ_{t:m}$:

$$\begin{aligned} V_\xi^{\pi,m}(æ_{<t}) &= (1-\gamma) \sum_{æ_{t:m}} \xi^\pi(æ_{t:m} \mid æ_{<t}) G_{t-1:m} \\ &= (1-\gamma) \sum_{æ_{t:m}} \left[\sum_\nu w(\nu \mid æ_{<t})\, \nu^\pi(æ_{t:m} \mid æ_{<t}) \right] G_{t-1:m} \\ &= \sum_\nu w(\nu \mid æ_{<t}) \underbrace{(1-\gamma) \sum_{æ_{t:m}} \nu^\pi(æ_{t:m} \mid æ_{<t}) G_{t-1:m}}_{= V_\nu^{\pi,m}(æ_{<t})} . \end{aligned}$$

In the last step we exchanged $\sum_{æ_{t:m}}$ and $\sum_\nu$; this is valid because both are sums of non-negative terms (or, for finite m, the sum over $æ_{t:m}$ is finite). So:

$$V_\xi^{\pi,m}(æ_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid æ_{<t})\, V_\nu^{\pi,m}(æ_{<t}).$$

Fact 4.1 (Dominated convergence for sums)**.** *If $a_{\nu,m} \to a_\nu$ as $m \to \infty$ for each ν, and $|a_{\nu,m}| \le b_\nu$ for all m with $\sum_\nu b_\nu < \infty$, then $\sum_\nu a_{\nu,m} \to \sum_\nu a_\nu$. (For finite sums this is trivial.)*

Exercise 4.2 (Infinite-horizon linearity). [17] (∗) Show that the result extends to $m = \infty$:

$$V_\xi^\pi(æ_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid æ_{<t})\, V_\nu^\pi(æ_{<t}).$$

Hint: Take $m \to \infty$ in the finite-horizon result. You will need Fact 4.1 to exchange limit and sum for countably infinite $\mathcal{M}$.

Solution to Exercise 4.2

The left side converges: $V_\xi^{\pi,m}(æ_{<t}) \to V_\xi^\pi(æ_{<t})$ by definition.
For the right side, we need to push $\lim_{m\to\infty}$ through $\sum_\nu$. If $\mathcal{M}$ is finite this is immediate. For countably infinite $\mathcal{M}$, we apply Fact 4.1 (dominated convergence for sums):

- For each ν: $a_{\nu,m} := w(\nu \mid æ_{<t})\, V_\nu^{\pi,m}(æ_{<t}) \to w(\nu \mid æ_{<t})\, V_\nu^\pi(æ_{<t})$ as $m \to \infty$ by definition.

- Domination: $|a_{\nu,m}| \le w(\nu \mid æ_{<t}) \cdot 1 =: b_\nu$ for all m, since $V_\nu^{\pi,m} \in [0,1]$ (Exercise 1.3).
- Summability: $\sum_\nu b_\nu = \sum_\nu w(\nu \mid æ_{<t}) = 1 < \infty$.

So Fact 4.1 gives:

$$V_\xi^\pi(æ_{<t}) \;=\; \lim_{m\to\infty} \sum_\nu w(\nu \mid æ_{<t})\, V_\nu^{\pi,m}(æ_{<t}) \;=\; \sum_\nu w(\nu \mid æ_{<t})\, V_\nu^\pi(æ_{<t}).$$

Exercise 4.1 shows V_ξ^π is linear in ν. The optimal value V_ξ^* is only convex: a single policy must perform well across all $\nu \in \mathcal{M}$ simultaneously, rather than being tailored to each ν individually.

Exercise 4.3. [10] $(*)$ (Convexity of V_ξ^*) Using Exercise 4.1, show that for finite m:

$$V_\xi^{*,m}(æ_{<t}) \;\le\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\, V_\nu^{*,m}(æ_{<t}).$$

Hint: Apply Exercise 4.1 with $\pi = \pi_\xi^*$, the Bayes-optimal policy for ξ.

Solution to Exercise 4.3

By Exercise 2.3, a deterministic Bayes-optimal policy π_ξ^* exists with $V_\xi^{\pi_\xi^*,m} = V_\xi^{*,m}$. Applying Exercise 4.1 with $\pi = \pi_\xi^*$:

$$V_\xi^{*,m}(æ_{<t}) \;=\; V_\xi^{\pi_\xi^*,m}(æ_{<t}) \;=\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\, V_\nu^{\pi_\xi^*,m}(æ_{<t}).$$

Since π_ξ^* is optimal for ξ but not necessarily for each individual ν, we have $V_\nu^{\pi_\xi^*,m}(æ_{<t}) \le V_\nu^{*,m}(æ_{<t})$ for each ν. Since $w(\nu \mid æ_{<t}) \ge 0$:

$$V_\xi^{*,m}(æ_{<t}) \;\le\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\, V_\nu^{*,m}(æ_{<t}).$$

Exercise 4.4. [15] $(*)$ (Non-linearity of V_ξ^*) Show by example that the inequality in Exercise 4.3 can be strict, i.e. $V_\xi^{*,m}$ is *not* linear in ν.

Hint: Consider $\mathcal{M} = \{\nu_0, \nu_1\}$: predicting a two-headed coin vs. a two-tailed coin.

Solution to Exercise 4.4

Coin-flip prediction. Let $\mathcal{A} = \mathcal{O} = \mathcal{R} = \{0,1\}$, $\mathcal{M} = \{\nu_0, \nu_1\}$ with $w_{\nu_0} = w_{\nu_1} = \frac{1}{2}$, horizon $m = 1$. Environment ν_i always shows outcome i, and the agent is rewarded for a correct prediction:

$$\nu_i(e_t \mid æ_{<t}\, a_t) \;:\; o_t = i, \quad r_t = [\![a_t = i]\!].$$

In ν_i the optimal policy plays $a_t = i$, achieving $V_{\nu_i}^{*,m} = 1$. In $\xi = \frac{1}{2}\nu_0 + \frac{1}{2}\nu_1$ the outcome is a fair coin flip, so no policy predicts better than chance: $V_{\xi}^{*,m} = \frac{1}{2}$. Therefore:

$$\sum_{\nu} w_\nu\, V_\nu^{*,m}(æ_{<t}) \;=\; \tfrac{1}{2}\cdot 1 + \tfrac{1}{2}\cdot 1 \;=\; 1 \;>\; \tfrac{1}{2} \;=\; V_\xi^{*,m}(æ_{<t}).$$

5. The Expectimax Form of AIXI

We have specified AIXI only implicitly, as the Bayes-optimal policy π_ξ^* (Definition 0.4). We now unroll this definition into the explicit **expectimax** expression: an alternating sequence of maximizations over actions and ξ-expectations over percepts.

Exercise 5.1 (Expectimax form of AIXI). **[20]** By iterating the finite-horizon Bellman optimality equation (Exercise 2.4), show that

$$\begin{aligned} V_\xi^{*,m}(æ_{<t}) \;=\; (1-\gamma) \max_{a_t} \sum_{e_t} \xi(e_t \mid æ_{<t} a_t) \\ \max_{a_{t+1}} \sum_{e_{t+1}} \xi(e_{t+1} \mid æ_{<t+1} a_{t+1}) \cdots \\ \cdots \max_{a_m} \sum_{e_m} \xi(e_m \mid æ_{<m} a_m)\, G_{t-1:m} \end{aligned}$$

and hence, collecting the percept factors with the chain rule (Exercise 0.2) and taking $m \to \infty$ (Exercises 2.5 and 2.7), that AIXI selects the action

$$\begin{aligned} a_t \;&=\; \pi_\xi^*(æ_{<t}) \\ &\in\; \arg\max_{a_t} \lim_{m\to\infty} \sum_{e_t} \max_{a_{t+1}} \sum_{e_{t+1}} \cdots \max_{a_m} \sum_{e_m} \xi(e_{t:m} \mid æ_{<t}\, a_{t:m})\, G_{t-1:m} \end{aligned}$$

Hint: Prove the first display by backward induction on t from m down to 1, using the finite-horizon Bellman optimality equation (Exercise 2.4) and the return recursion $G_{t-1:m} = r_t + \gamma\, G_{t:m}$ (Definition 0.3). At each step the leftover reward

term $(1-\gamma)\, r_t$ is constant in the deeper variables; carry it inward using $\sum_{e_k} \xi(e_k \mid æ_{<k} a_k) = 1$ and $c + \max(\cdot) = \max(c + \cdot)$.

Solution to Exercise 5.1

Abbreviate the one-step predictor $\xi_k := \xi(e_k \mid æ_{<k} a_k)$, and write the **expectimax operator** over steps $i, \dots, j$ as

$$\max_{i:j} \;:=\; \max_{a_i} \sum_{e_i} \xi_i \max_{a_{i+1}} \sum_{e_{i+1}} \xi_{i+1} \cdots \max_{a_j} \sum_{e_j} \xi_j,$$

the alternating "maximize over the action, average over the percept" chain, with the ξ_k weights built in; the single step is $\max_k := \max_{a_k} \sum_{e_k} \xi_k$. Two facts:

- **(E1) Composition:** $\max_{i:j} = \max_i \max_{i+1:j}$, by nesting the definition.
- **(E2) Affine pass-through:** for any $c \in \mathbb{R}$ and $\lambda \geq 0$ not depending on $(a_i, e_i, \dots, a_j, e_j)$,

  $$c + \lambda \max_{i:j} X \;=\; \max_{i:j} (c + \lambda X),$$

  since each layer has $\sum_{e_k} \xi_k = 1$ (so $\sum_{e_k} \xi_k (c + \lambda Y) = c + \lambda \sum_{e_k} \xi_k Y$) and $\max_a (c + \lambda\, g(a)) = c + \lambda \max_a g(a)$ for $\lambda \geq 0$.

We prove by backward induction on $t = m, m-1, \dots, 1$ that

$$V_\xi^{*,m}(æ_{<t}) \;=\; (1-\gamma) \max_{t:m} G_{t-1:m}.$$

Base case $t = m$. Here $G_{m-1:m} = r_m$ and $V_\xi^{*,m}(æ_{1:m}) = 0$ (the return past the horizon is empty, $G_{m:m} = 0$). The Bellman optimality equation (Exercise 2.4) for $\nu = \xi$ gives

$$V_\xi^{*,m}(æ_{<m}) \;=\; \max_{a_m} \sum_{e_m} \xi_m \big[(1-\gamma)\, r_m + \gamma \cdot 0\big] \;=\; (1-\gamma) \max_{m:m} G_{m-1:m}.$$

Inductive step. Assume the claim at time $t+1$, i.e. $V_\xi^{*,m}(æ_{1:t}) = (1-\gamma) \max_{t+1:m} G_{t:m}$ for every length-t history (recall $æ_{<t+1} = æ_{1:t}$). The Bellman optimality equation (Exercise 2.4) at time t reads

$$V_\xi^{*,m}(æ_{<t}) \;=\; \max_t \big[(1-\gamma)\, r_t + \gamma\, V_\xi^{*,m}(æ_{1:t})\big].$$

Substituting the inductive hypothesis and pulling out $(1-\gamma)$,

$$V_\xi^{*,m}(æ_{<t}) \;=\; (1-\gamma)\,\max_{t}\sum\,\big[r_t+\gamma\,\max_{t+1:m}\sum\,G_{t:m}\big].$$

As r_t and γ do not depend on $(a_{t+1},e_{t+1},\dots,a_m,e_m)$, property (E2) carries them inside the inner operator, and the return recursion $G_{t-1:m}=r_t+\gamma\,G_{t:m}$ (Definition 0.3) gives

$$r_t+\gamma\,\max_{t+1:m}\sum\,G_{t:m}\;\overset{(E2)}{=}\;\max_{t+1:m}\sum\,(r_t+\gamma\,G_{t:m})\;=\;\max_{t+1:m}\sum\,G_{t-1:m}.$$

Recombining the two operators by (E1),

$$V_\xi^{*,m}(æ_{<t}) \;=\; (1-\gamma)\,\max_{t}\sum\,\max_{t+1:m}\sum\,G_{t-1:m}\;=\;(1-\gamma)\,\max_{t:m}\sum\,G_{t-1:m},$$

which expands back into the max / $\sum$ layers of the first display.

Infinite horizon and the action. Taking $m\to\infty$ (Exercises 2.5 and 2.7), $G_{t-1:m}\to G_{\geq t-1}$ and the chain extends indefinitely, so $V_\xi^*(æ_{<t})=(1-\gamma)\max_{t:\infty}\sum G_{\geq t-1}$. Collecting the per-step factors by the chain rule (Exercise 0.2, iterated), $\prod_{k=t}^{m}\xi_k=\xi(e_{t:m}\mid æ_{<t}\,a_{t:m})$. A Bayes-optimal action maximizes this expression; peeling the outer $\max_{a_t}$ off as an arg max and dropping the positive constant $(1-\gamma)$ (which does not move the maximizer) gives the stated action.

On-Policy Value Convergence

The following problems prove the first two main results: on-policy value convergence (Section 7), and that AIXI can't be fooled in deterministic environments (Section 8). The path to the self-optimizing property (advanced stretch goal) resumes at Section 9.

6. Bounding Expectation Differences by Total Variation

Definition 6.1 (Total variation distance)**.** The **total variation distance** between probability measures P and Q on a countable set Ω is $\mathrm{TV}_\Omega(P,Q):=\sup_{S\subseteq\Omega}|P(S)-Q(S)|$, where $P(S):=\sum_{\omega\in S}P(\omega)$.

Definition 6.2 (Expectation under P)**.** For a probability measure P on a countable set Ω and a function $f : \Omega \to \mathbb{R}$, the **expectation** of f under P is $\mathbb{E}_P[f] := \sum_{\omega \in \Omega} f(\omega)\, P(\omega)$.

The following technical lemma is needed for the on-policy value convergence proof.

Exercise 6.1. **[15]** $(*)$ Let $f : \Omega \to [0, c]$. Show that $\big|\mathbb{E}_P[f] - \mathbb{E}_Q[f]\big| \leq c \cdot \mathrm{TV}_\Omega(P, Q)$.

Hint: Define $A^+ = \{\omega \in \Omega : P(\omega) \geq Q(\omega)\}$ and decompose the expectation difference as a sum over A^+ and its complement $A^- = \Omega \setminus A^+$.

Solution to Exercise 6.1

Write out the expectation difference using Definition 6.2:

$$\mathbb{E}_P[f] - \mathbb{E}_Q[f] \;=\; \sum_{\omega \in \Omega} f(\omega)\big(P(\omega) - Q(\omega)\big).$$

Define $A^+ := \{\omega \in \Omega : P(\omega) \geq Q(\omega)\}$ and $A^- := \Omega \setminus A^+$, and split the sum:

$$\mathbb{E}_P[f] - \mathbb{E}_Q[f] \;=\; \sum_{\omega \in A^+} f(\omega)\big(P(\omega) - Q(\omega)\big) \;+\; \sum_{\omega \in A^-} f(\omega)\big(P(\omega) - Q(\omega)\big).$$

Bounding the sum over A^+. On A^+, $P(\omega) - Q(\omega) \geq 0$ and $f(\omega) \leq c$, so:

$$\begin{aligned}
\sum_{\omega \in A^+} f(\omega)\big(P(\omega) - Q(\omega)\big) \;&\leq\; c \sum_{\omega \in A^+} \big(P(\omega) - Q(\omega)\big) \\
&=\; c \cdot \big(P(A^+) - Q(A^+)\big) \\
&\leq\; c \cdot \sup_S |P(S) - Q(S)| \\
&=\; c \cdot \mathrm{TV}_\Omega(P, Q).
\end{aligned}$$

Bounding the sum over A^-. On A^-, $P(\omega) - Q(\omega) < 0$ and $f(\omega) \geq 0$, so every term $f(\omega)(P(\omega) - Q(\omega)) \leq 0$:

$$\sum_{\omega \in A^-} f(\omega)\big(P(\omega) - Q(\omega)\big) \;\leq\; 0.$$

Combining. $\mathbb{E}_P[f] - \mathbb{E}_Q[f] \leq c \cdot \mathrm{TV}_\Omega(P, Q) + 0 = c \cdot \mathrm{TV}_\Omega(P, Q)$.
The other direction. Swapping P and Q: define $\tilde{A}^+ = \{\omega : Q(\omega) \geq P(\omega)\} = A^-$. The same argument gives $\mathbb{E}_Q[f] - \mathbb{E}_P[f] \leq c \cdot \mathrm{TV}_\Omega(Q, P) = c \cdot \mathrm{TV}_\Omega(P, Q)$.
Therefore $|\mathbb{E}_P[f] - \mathbb{E}_Q[f]| \leq c \cdot \mathrm{TV}_\Omega(P, Q)$.

7. On-Policy Value Convergence of Bayes

The following definitions are needed for the on-policy value convergence theorem.

Definition 7.1 (Probability of events)**.** A **finite-length event** is a set $A \subseteq (\mathcal{A} \times \mathcal{E})^t$ of histories of fixed length t. Its probability under ν^π is

$$\nu^\pi(A) \;:=\; \sum_{æ_{1:t} \in A} \nu^\pi(æ_{1:t}).$$

An **event on infinite histories** is any set built from finite-length events by countable unions, intersections, and complements.[a] Probabilities of such events are uniquely determined by two properties:

- **Normalization:** $\nu^\pi\big((\mathcal{A} \times \mathcal{E})^\infty\big) = 1$.
- **Countable additivity:** if $A_1, A_2, \ldots$ are pairwise disjoint events, then $\nu^\pi\big(\bigcup_{n=1}^\infty A_n\big) = \sum_{n=1}^\infty \nu^\pi(A_n)$.

From these, all standard rules of probability can be derived (e.g. $\nu^\pi(A \cup B) = \nu^\pi(A) + \nu^\pi(B) - \nu^\pi(A \cap B)$, etc.), though we will not prove them here. Two derived properties we use explicitly:

- **Complement:** $\nu^\pi(A^c) = 1 - \nu^\pi(A)$.
- **Monotone limits:** if $A_1 \supseteq A_2 \supseteq \cdots$, then $\nu^\pi\big(\bigcap_{n=1}^\infty A_n\big) = \lim_{n\to\infty} \nu^\pi(A_n)$.

Further useful notions:

- **Conditional probability:** The probability of event A given observed history $æ_{<t}$ is

$$\nu^\pi(A \mid æ_{<t}) \;:=\; \frac{\nu^\pi(\{æ_{<t}h : h \in A\})}{\nu^\pi(æ_{<t})}.$$

[a] We are deliberately avoiding a formal treatment of measure theory here. Not all subsets of $(\mathcal{A} \times \mathcal{E})^\infty$ are measurable; we restrict to "nice" (measurable) sets built from finite-prefix conditions via countable set operations, which suffice for everything in this sheet. For a rigorous treatment using σ-algebras and probability measures, see [HQC24, Chapter 2.2].

Example 7.2 (A fair coin shows 1 infinitely often)**.** Let $\mathcal{A} = \mathcal{O} = \{0, 1\}$, $\mathcal{R} = \{0\}$, and let ν be a fair coin that ignores the action: $\nu(o_t = 1 \mid æ_{<t}\, a_t) = \frac{1}{2}$ for all $æ_{<t}, a_t$. Consider the event $F :=$ "$o_t = 1$ for infinitely many t".

Step 1: Build F^c from finite-prefix events. For each t, the set $\{o_t = 0\} := \{æ_{1:t} :$

$o_t = 0\}$ is a finite-length event. For each N and $T \geq N$, the set $D_{N,T} := \bigcap_{t=N}^{T}\{o_t = 0\}$ is also a finite-length event (determined by time T). Define $C_N := \bigcap_{t=N}^{\infty}\{o_t = 0\}$ ("all zeros from time N onwards"): this is an infinite-history event, built as a countable intersection of finite-length events. Then $F^c = \bigcup_{N=1}^{\infty} C_N$ ("eventually all zeros").

Step 2: Compute $\nu^\pi(C_N) = 0$. Since $D_{N,T} \supseteq D_{N,T+1} \supseteq \cdots$ (adding more constraints), the monotone limit property (for decreasing sets) gives:

$$\nu^\pi(C_N) \;=\; \lim_{T\to\infty} \nu^\pi(D_{N,T}) \;=\; \lim_{T\to\infty} \left(\tfrac{1}{2}\right)^{T-N+1} \;=\; 0.$$

(The probability $\nu^\pi(D_{N,T}) = (1/2)^{T-N+1}$ is independent of π, since ν ignores actions.)

Step 3: By countable additivity: $\nu^\pi(F^c) \leq \sum_{N=1}^{\infty} \nu^\pi(C_N) = 0$, so $\nu^\pi(F) = 1$. A fair coin shows 1 infinitely often, almost surely.
As such, F^c (the set of all infinite histories containing only finitely many 1s) is a ν^π-measure-zero set. Such histories "exist" as infinite sequences, but occur with probability zero.

Definition 7.3 (Covering[1])**.** We say Q **covers** P if Q is positive everywhere P is: $P(A) > 0 \implies Q(A) > 0$ for all events A. Equivalently: any event that Q rules out, P also rules out. By Section 3, ξ^π covers μ^π.

Definition 7.4 (Convergence ν^π-almost surely)**.** Let $f_t : \mathcal{H}^{t-1} \to \mathbb{R}$ be a sequence of functions, where f_t depends on the history $æ_{<t}$ of length $t-1$. We write $f_t(æ_{<t}) \to 0$ ν^π**-almost surely** (ν^π-a.s.) if

$$\nu^\pi\Big(\Big\{æ_{1:\infty} : f_t(æ_{<t}) \not\to 0\Big\}\Big) = 0.$$

This set is built from finite-prefix conditions: $\{f_t \not\to 0\} = \bigcup_{n=1}^{\infty}\bigcap_{N=1}^{\infty}\bigcup_{t=N}^{\infty}\{|f_t(æ_{<t})| > \frac{1}{n}\}$, so its probability is well-defined (Definition 7.1).

Example 7.5 (Unpacking "$f_t \not\to 0$")**.** The set $\{f_t \not\to 0\}$ looks intimidating, but it reads naturally from the inside out:

- $\{|f_t(æ_{<t})| > \frac{1}{n}\}$ is a finite-length event: "at time t, the function is at least $\frac{1}{n}$ away from zero."

[1]The standard name is *absolute continuity* of P with respect to Q, written $P \ll Q$.

- $\bigcup_{t=N}^{\infty}\{|f_t| > \frac{1}{n}\}$: "at *some* time $t \geq N$, f_t is at least $\frac{1}{n}$ away from zero."
- $\bigcap_{N=1}^{\infty}\bigcup_{t=N}^{\infty}\{|f_t| > \frac{1}{n}\}$: "for *every* N, there is some $t \geq N$ where f_t is at least $\frac{1}{n}$ from zero", i.e., f_t exceeds $\frac{1}{n}$ *infinitely often.*
- $\bigcup_{n=1}^{\infty}\bigcap_{N=1}^{\infty}\bigcup_{t=N}^{\infty}\{|f_t| > \frac{1}{n}\}$: "for *some* $\frac{1}{n} > 0$, f_t exceeds $\frac{1}{n}$ infinitely often."

This last condition is exactly $\{f_t \not\to 0\}$: convergence $f_t \to 0$ means that for *every* $\varepsilon > 0$, $|f_t| \leq \varepsilon$ for all sufficiently large t. Its negation is that *some* $\varepsilon > 0$ is exceeded infinitely often.
Each layer is a countable union or intersection of the previous, so the whole set is a well-defined event on infinite histories.

The Bayesian agent uses the mixture ξ because the true environment μ is unknown. A natural question: does planning with ξ eventually become as good as planning with μ? The following theorem says *yes*: the value of any fixed policy π, as evaluated by ξ, converges to the value under the true environment μ, along histories that μ^π actually generates.[2] This tells us that the Bayesian mixture "learns" to predict the true environment's value, on policy.

Theorem 7.6 (On-policy value convergence; [HQC24, Theorem 7.3.1])**.** *For any $\mu \in \mathcal{M}$ and any policy π: $V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t}) \to 0$ as $t \to \infty$, μ^π-almost surely. That is, the set of infinite histories along which the value difference does not vanish has μ^π-probability zero:*

$$\mu^\pi\Big(\Big\{æ_{1:\infty} : V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t}) \not\to 0\Big\}\Big) = 0.$$

The proof reduces to a finite-horizon TV bound (Exercise 7.1), a covering argument (Exercise 7.2), and the Blackwell–Dubins theorem (Exercise 7.3). The only ingredient we do not prove is Blackwell–Dubins itself, which we state as a given fact.

Exercise 7.1 (Finite-horizon TV bound on value difference). **[10]**
Using Section 6, show that for finite m:

$$\big|V_\xi^{\pi,m}(æ_{<t}) - V_\mu^{\pi,m}(æ_{<t})\big| \;\leq\; \mathrm{TV}_{\mathcal{H}^{m-t+1}}\big(\xi^\pi(\cdot \mid æ_{<t}),\; \mu^\pi(\cdot \mid æ_{<t})\big),$$

where $\mathcal{H}^{m-t+1} := (\mathcal{A} \times \mathcal{E})^{m-t+1}$ is the set of all future history segments $æ_{t:m}$.

[2] If the true environment is μ then we don't care about the behaviour of the Bayesian agent on histories that have μ^π-probability zero: such histories will never be observed anyway.

Solution to Exercise 7.1

We want to apply Section 6. Identify:

- $\Omega := \mathcal{H}^{m-t+1} = (\mathcal{A} \times \mathcal{E})^{m-t+1}$, the finite set of future history segments $æ_{t:m}$.
- $P := \mu^\pi(\cdot \mid æ_{<t})$ and $Q := \xi^\pi(\cdot \mid æ_{<t})$, the conditional measures over Ω.
- $f(æ_{t:m}) := (1-\gamma)G_{t-1:m}$, which satisfies $f \in [0,1]$ by Exercise 1.3 (so $c = 1$).

Then $V_\nu^{\pi,m}(æ_{<t}) = \sum_{æ_{t:m}} \nu^\pi(æ_{t:m} \mid æ_{<t})\, f(æ_{t:m})$ is exactly $\mathbb{E}_P[f]$ (for $\nu = \mu$) or $\mathbb{E}_Q[f]$ (for $\nu = \xi$). Applying Section 6:

$$\left|V_\xi^{\pi,m}(æ_{<t}) - V_\mu^{\pi,m}(æ_{<t})\right| \;=\; \left|\mathbb{E}_Q[f] - \mathbb{E}_P[f]\right| \;\le\; 1 \cdot \mathrm{TV}_{\mathcal{H}^{m-t+1}}(P,Q).$$

Exercise 7.2 (Covering). **[05]** Show that ξ^π covers μ^π (Definition 7.3).

Hint: Use Section 3.

Solution to Exercise 7.2

By Section 3, $\xi^\pi(æ_{1:m}) \ge w_\mu \cdot \mu^\pi(æ_{1:m})$ for all histories and all m. So for any event A: $\mu^\pi(A) > 0 \implies \xi^\pi(A) \ge w_\mu \cdot \mu^\pi(A) > 0$. Hence ξ^π covers μ^π.

To complete the proof, we need the TV distance on finite segments to vanish μ^π-a.s. as $t \to \infty$. This follows from the following deep result, which we state without proof:

Fact 7.7 (Blackwell–Dubins merging of opinions [BD62])**.** *If Q covers P, then $\sup_S |P(S \mid æ_{<t}) - Q(S \mid æ_{<t})| \to 0$ P-almost surely, where the supremum ranges over all measurable events S (including events on infinite histories).*

The proof of Blackwell–Dubins requires the Radon–Nikodym derivative and Lévy's martingale convergence theorem: tools from measure theory that are beyond the scope of this sheet. See [HQC24, Chapter 3.9] for discussion.

Exercise 7.3 (On-policy convergence). **[15]** Using Fact 7.7, Exercise 7.1 and Exercise 7.2, prove Theorem 7.6.

Solution to Exercise 7.3

Step 1: Finite-horizon bound. By Exercise 7.1, for each finite m:

$$\begin{aligned}\left|V_\xi^{\pi,m}(æ_{<t}) - V_\mu^{\pi,m}(æ_{<t})\right| &\le \mathrm{TV}_{\mathcal{H}^{m-t+1}}\big(\xi^\pi(\cdot \mid æ_{<t}),\, \mu^\pi(\cdot \mid æ_{<t})\big) \\ &\le \sup_S \left|\xi^\pi(S \mid æ_{<t}) - \mu^\pi(S \mid æ_{<t})\right|,\end{aligned}$$

where the second inequality uses that the sup over subsets of $\mathcal{H}^{m-t+1}$ is at most the sup over all measurable events.
Step 2: Take $m \to \infty$. The LHS converges to $|V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t})|$ (by definition of the infinite-horizon value as the pointwise limit). The RHS does not depend on m. Hence

$$\left|V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t})\right| \le \sup_S \left|\xi^\pi(S \mid æ_{<t}) - \mu^\pi(S \mid æ_{<t})\right|.$$

Step 3: Take $t \to \infty$. By Exercise 7.2, ξ^π covers μ^π. By Fact 7.7 (Blackwell–Dubins) with $P = \mu^\pi$ and $Q = \xi^\pi$, the RHS tends to 0 as $t \to \infty$, μ^π-a.s. Therefore

$$\left|V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t})\right| \to 0 \qquad \mu^\pi\text{-a.s.}$$

Exercise 7.4. **[40]** $(*)$ Prove Fact 7.7.

Hint: See [BD62]. The proof uses the Radon–Nikodym derivative dP/dQ, Lévy's martingale convergence theorem, and the Lebesgue decomposition. No elementary proof is known that avoids graduate-level measure theory.

8. AIXI Cannot Be Fooled in Deterministic Environments

Theorem 8.1 (AIXI cannot act poorly in good environments)**.** *If $\mu \in \mathcal{M}$ is deterministic and $V_\mu^*(æ_{<t}) > \varepsilon > 0$ for all t along the history generated by μ and π_ξ^*, then $V_\xi^*(æ_{<t}) \ge w_\mu\, \varepsilon > 0$ for all t. [HQC24, Theorem 7.4.10]*

Exercise 8.1. **[10]** Show that $V_\xi^*(æ_{<t}) \ge w(\mu \mid æ_{<t}) \cdot V_\mu^*(æ_{<t})$.

Hint: Use linearity of V^π (Exercise 4.2).

Solution to Exercise 8.1

By definition, $V_\xi^*(æ_{<t}) = \sup_\pi V_\xi^\pi(æ_{<t}) \geq V_\xi^{\pi'}(æ_{<t})$ for any policy π'. By Exercise 4.2 (linearity of V^π in the environment):

$$V_\xi^{\pi'}(æ_{<t}) \;=\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\, V_\nu^{\pi'}(æ_{<t}).$$

Choose $\pi' = \pi_\mu^*$, the deterministic optimal policy for μ whose existence is guaranteed by Exercise 2.7:

$$V_\xi^*(æ_{<t}) \;\geq\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\, V_\nu^{\pi_\mu^*}(æ_{<t}).$$

Every term is non-negative ($w(\nu \mid æ_{<t}) \geq 0$ and $V_\nu^{\pi_\mu^*} \geq 0$). Dropping all terms except $\nu = \mu$:

$$V_\xi^*(æ_{<t}) \;\geq\; w(\mu \mid æ_{<t})\, V_\mu^{\pi_\mu^*}(æ_{<t}) \;=\; w(\mu \mid æ_{<t})\, V_\mu^*(æ_{<t}).$$

Exercise 8.2. **[15]** Show $w(\mu \mid æ_{<t}) \geq w_\mu$ when μ is deterministic, and combine with Exercise 8.1 to prove Theorem 8.1.

Solution to Exercise 8.2

From Definition 0.1, the posterior weight is:

$$w(\mu \mid æ_{<t}) \;=\; w_\mu \cdot \frac{\mu(æ_{<t})}{\xi(æ_{<t})} \;=\; w_\mu \cdot \frac{\prod_{k=1}^{t-1} \mu(e_k \mid æ_{<k}a_k)}{\prod_{k=1}^{t-1} \xi(e_k \mid æ_{<k}a_k)}.$$

Since μ is deterministic, for the history that μ actually generates, $\mu(e_k \mid æ_{<k}a_k) = 1$ for each k (the environment produces each percept with certainty). So:

$$\mu(æ_{<t}) \;=\; \prod_{k=1}^{t-1} 1 \;=\; 1.$$

Meanwhile, $\xi(e_k \mid æ_{<k}a_k) \leq 1$ for each k (it is a probability), so $\xi(æ_{<t}) = \prod_{k=1}^{t-1} \xi(e_k \mid æ_{<k}a_k) \leq 1$.
Therefore $w(\mu \mid æ_{<t}) = w_\mu \cdot 1/\xi(æ_{<t}) \geq w_\mu$.
Combining with Exercise 8.1:

$$V_\xi^*(æ_{<t}) \;\geq\; w(\mu \mid æ_{<t}) \cdot V_\mu^*(æ_{<t}) \;\geq\; w_\mu \cdot V_\mu^*(æ_{<t}) \;>\; w_\mu \cdot \varepsilon \;>\; 0.$$

Remark

With the Solomonoff prior, $w_\mu = 2^{-K(\mu)}$ may be astronomically small. The result guarantees non-zero value, not near-optimal value.

Sections 9–11: Self-Optimizing Policy (Finite Model Class)

We now prove: if *any* policy can learn to act optimally, π^*_ξ also learns. We restrict to finite $\mathcal{M} = \{\nu_1, \ldots, \nu_N\}$ where each ν_i is a proper probability measure.

9. Likelihood Ratios Are Martingales

The following definition and theorem state the main goal of Sections 9–11.

Definition 9.1 (Self-optimizing)**.** Fix a historic policy π. A policy $\tilde{\pi}$ is **self-optimizing** for $\mathcal{M}$ with respect to π if for every $\nu \in \mathcal{M}$: $V^*_\nu(æ_{<t}) - V^{\tilde{\pi}}_\nu(æ_{<t}) \to 0$ as $t \to \infty$, ν^π-almost surely.
The percepts $e_{1:\infty}$ are sampled from ν; the historic actions $a_{<t}$ come from π, which may differ from $\tilde{\pi}$.

Theorem 9.2 (Self-optimizing; [HQC24, Theorem 7.5.2])**.** *Let $\mathcal{M}$ be finite and fix a historic policy π. If there exists a $\tilde{\pi}$ self-optimizing for $\mathcal{M}$ with respect to π, then π^*_ξ is also self-optimizing for $\mathcal{M}$ with respect to π: for any $\mu \in \mathcal{M}$, $V^*_\mu(æ_{<t}) - V^{\pi^*_\xi}_\mu(æ_{<t}) \to 0$ μ^π-almost surely.*

Definition 9.3 (Supermartingale)**.** A sequence of functions $X_t : \mathcal{H}^t \to \mathbb{R}_{\geq 0}$ (each depending on the history $æ_{1:t}$) is a μ^π-**supermartingale** if $\mathbb{E}_{\mu^\pi}[X_t \mid æ_{<t}] \leq X_{t-1}$ for all t; that is,

$$\sum_{æ_t} \pi(a_t \mid æ_{<t})\, \mu(e_t \mid æ_{<t}a_t)\, X_t(æ_{1:t}) \;\leq\; X_{t-1}(æ_{<t}).$$

If equality holds, it is a μ^π-**martingale**.

Intuition. A supermartingale is a quantity whose expected future value, given the present, is no larger than its current value: it cannot "drift upward on average." A martingale is the equality case: expected future value equals current value. The key example is the **likelihood ratio** $X_{\nu,t}(æ_{1:t}) := \nu(æ_{1:t})/\mu(æ_{1:t})$: how much more (or less) likely the observed history is under ν than under the true environment μ. Under μ, this

ratio is a martingale: the true environment does not, on average, favour any alternative ν over itself. Non-negativity plus the (super)martingale structure forces $X_{\nu,t}$ to converge (by Doob's theorem below), which is the engine behind the self-optimizing proof: it lets us separate environments that remain plausible ($X_{\nu,\infty} > 0$) from those that are eventually ruled out ($X_{\nu,\infty} = 0$), and handle each case differently in Section 11.

Fact 9.4 (Doob's supermartingale convergence)**.** *If $(X_t)_{t\geq 0}$ is a non-negative μ^π-supermartingale, then there exists a function $X_\infty : \mathcal{H}^\infty \to \mathbb{R}_{\geq 0}$ of the infinite history such that $X_t(æ_{1:t}) \to X_\infty(æ_{1:\infty})$ as $t \to \infty$, with $X_\infty(æ_{1:\infty}) < \infty$, for μ^π-almost every infinite history $æ_{1:\infty}$. That is:*

$$\mu^\pi\Big(\Big\{æ_{1:\infty} \in (\mathcal{A}\times\mathcal{E})^\infty : X_t(æ_{1:t}) \not\to X_\infty(æ_{1:\infty})\Big\}\Big) \;=\; 0.$$

For each $\nu \in \mathcal{M}$, define the likelihood ratio

$$X_{\nu,t} : \mathcal{H}^t \to \mathbb{R}_{\geq 0}, \qquad X_{\nu,t}(æ_{1:t}) \;:=\; \frac{\nu(æ_{1:t})}{\mu(æ_{1:t})}, \qquad X_{\nu,0} := 1.$$

$X_{\nu,t}$ is a *function* of the history $æ_{1:t}$, not a number: different histories give different values. Throughout this section and the next, any unqualified statement of the form "$X_{\nu,t} \geq c$" or "$X_{\nu,t} \to X_{\nu,\infty}$" is shorthand for "$X_{\nu,t}(æ_{1:t}) \geq c$" or "$X_{\nu,t}(æ_{1:t}) \to X_{\nu,\infty}(æ_{1:\infty})$" holding μ^π-almost surely, i.e. on every infinite history outside a set of μ^π-measure zero. We will not track these null sets explicitly; their countable union over all claims is still a μ^π-null set.

Exercise 9.1. **[15]** Show that $X_{\nu,t}$ is a μ^π-martingale: $\mathbb{E}_{\mu^\pi}[X_{\nu,t} \mid æ_{<t}] = X_{\nu,t-1}$.

Hint: Write $X_{\nu,t}(æ_{1:t}) = X_{\nu,t-1}(æ_{<t}) \cdot \nu(e_t \mid æ_{<t}a_t)/\mu(e_t \mid æ_{<t}a_t)$. Sum over a_t, e_t; μ cancels, leaving $\sum_{e_t} \nu(e_t \mid æ_{<t}a_t) = 1$.

Solution to Exercise 9.1

We need to compute $\mathbb{E}_{\mu^\pi}[X_{\nu,t} \mid æ_{<t}]$. Conditioning on $æ_{<t}$ means a_t and e_t are the random quantities (sampled from π and μ respectively). First, write $X_{\nu,t}$ pointwise as a function of history in terms of $X_{\nu,t-1}$:

$$\begin{aligned} X_{\nu,t}(æ_{1:t}) \;&=\; \frac{\nu(æ_{1:t})}{\mu(æ_{1:t})} \;=\; \frac{\nu(æ_{<t})}{\mu(æ_{<t})} \cdot \frac{\nu(e_t \mid æ_{<t}a_t)}{\mu(e_t \mid æ_{<t}a_t)} \\ &=\; X_{\nu,t-1}(æ_{<t}) \cdot \frac{\nu(e_t \mid æ_{<t}a_t)}{\mu(e_t \mid æ_{<t}a_t)}, \end{aligned}$$

using the chain rule (Exercise 0.2) for both ν and μ. Now take the conditional

expectation. Since $X_{\nu,t-1}$ depends only on $æ_{<t}$, it comes out of the expectation:

$$\begin{aligned}\mathbb{E}_{\mu^\pi}[X_{\nu,t} \mid æ_{<t}] &= X_{\nu,t-1} \cdot \mathbb{E}_{\mu^\pi}\left[\frac{\nu(e_t \mid æ_{<t}a_t)}{\mu(e_t \mid æ_{<t}a_t)} \;\middle|\; æ_{<t}\right] \\ &= X_{\nu,t-1} \sum_{a_t \in \mathcal{A}} \pi(a_t \mid æ_{<t}) \sum_{e_t \in \mathcal{E}} \mu(e_t \mid æ_{<t}a_t) \cdot \frac{\nu(e_t \mid æ_{<t}a_t)}{\mu(e_t \mid æ_{<t}a_t)} \\ &= X_{\nu,t-1} \sum_{a_t} \pi(a_t \mid æ_{<t}) \sum_{e_t} \nu(e_t \mid æ_{<t}a_t).\end{aligned}$$

In the last step, $\mu(e_t \mid æ_{<t}a_t)$ cancels. Since ν is a probability measure, $\sum_{e_t} \nu(e_t \mid æ_{<t}a_t) = 1$ for every $æ_{<t}, a_t$; and $\sum_{a_t} \pi(a_t \mid æ_{<t}) = 1$. Therefore:

$$\mathbb{E}_{\mu^\pi}[X_{\nu,t} \mid æ_{<t}] \;=\; X_{\nu,t-1} \cdot 1 \cdot 1 \;=\; X_{\nu,t-1}.$$

Since $X_{\nu,t} \geq 0$ (a ratio of non-negative quantities), $(X_{\nu,t})_{t \geq 0}$ is a non-negative μ^π-martingale, in particular, a non-negative μ^π-supermartingale.

Exercise 9.2. **[05]** Apply Fact 9.4 to conclude $X_{\nu,t} \to X_{\nu,\infty} < \infty$ μ^π-a.s.

Solution to Exercise 9.2

$X_{\nu,t}$ is a non-negative μ^π-supermartingale by Exercise 9.1. By Fact 9.4 (Doob's convergence theorem), $X_{\nu,t}$ converges μ^π-a.s. to a finite limit: $X_{\nu,t} \to X_{\nu,\infty} < \infty$.

Exercise 9.3. **[10]** Define $X_{\xi,t} := \xi(æ_{1:t})/\mu(æ_{1:t})$. Show $X_{\xi,t} = \sum_\nu w_\nu X_{\nu,t}$ and $X_{\xi,t} \geq w_\mu > 0$.

Solution to Exercise 9.3

From the definition:

$$X_{\xi,t} \;:=\; \frac{\xi(æ_{1:t})}{\mu(æ_{1:t})} \;=\; \frac{\sum_\nu w_\nu \nu(æ_{1:t})}{\mu(æ_{1:t})} \;=\; \sum_\nu w_\nu \cdot \frac{\nu(æ_{1:t})}{\mu(æ_{1:t})} \;=\; \sum_\nu w_\nu X_{\nu,t}.$$

By Section 3: $\xi(æ_{1:t}) \geq w_\mu \cdot \mu(æ_{1:t})$, so $X_{\xi,t} = \xi(æ_{1:t})/\mu(æ_{1:t}) \geq w_\mu > 0$.
Since $\mathcal{M}$ is finite, $X_{\xi,t} = \sum_\nu w_\nu X_{\nu,t}$ is a finite sum of convergent sequences, so $X_{\xi,t} \to X_{\xi,\infty} := \sum_\nu w_\nu X_{\nu,\infty} < \infty$ μ^π-a.s., with $X_{\xi,\infty} \geq w_\mu > 0$.

10. Change of Measure

Exercise 10.1. [15] Let A be a set of finite histories of length m. Show:

$$\nu^\pi[æ_{1:m} \in A] \;=\; \mathbb{E}_{\mu^\pi}\big[X_{\nu,m} \cdot [\![æ_{1:m} \in A]\!]\big].$$

Hint: Use Exercise 0.1 to show $\nu^\pi/\mu^\pi = \nu/\mu = X_{\nu,m}$.

Solution to Exercise 10.1

By Exercise 0.1, $\nu^\pi(æ'_{1:m}) = \pi(æ'_{1:m}) \cdot \nu(æ'_{1:m})$ and $\mu^\pi(æ'_{1:m}) = \pi(æ'_{1:m}) \cdot \mu(æ'_{1:m})$. For any history $æ'_{1:m}$ with $\mu^\pi(æ'_{1:m}) > 0$, the policy factors cancel:

$$\begin{aligned}\nu^\pi(æ'_{1:m}) \;&=\; \frac{\nu^\pi(æ'_{1:m})}{\mu^\pi(æ'_{1:m})} \cdot \mu^\pi(æ'_{1:m}) \;=\; \frac{\nu(æ'_{1:m})}{\mu(æ'_{1:m})} \cdot \mu^\pi(æ'_{1:m}) \\ &=\; X_{\nu,m}(æ'_{1:m}) \cdot \mu^\pi(æ'_{1:m}).\end{aligned}$$

For histories with $\mu^\pi(æ'_{1:m}) = 0$: since $\mu^\pi = \pi \cdot \mu$, some $\pi(a_k \mid æ'_{<k}) = 0$, which forces $\nu^\pi(æ'_{1:m}) = 0$ too (the same π-factor appears in $\nu^\pi = \pi \cdot \nu$). So both sides are zero.

Summing over $æ'_{1:m} \in A$:

$$\begin{aligned}\nu^\pi[æ_{1:m} \in A] \;&=\; \sum_{æ'_{1:m} \in A} \nu^\pi(æ'_{1:m}) \;=\; \sum_{æ'_{1:m} \in A} X_{\nu,m}(æ'_{1:m}) \cdot \mu^\pi(æ'_{1:m}) \\ &=\; \mathbb{E}_{\mu^\pi}\big[X_{\nu,m} \cdot [\![æ_{1:m} \in A]\!]\big].\end{aligned}$$

Exercise 10.2. (Given.) The identity extends to infinite histories: for any event $A \subseteq (\mathcal{A} \times \mathcal{E})^\infty$,

$$\nu^\pi[A] \;=\; \mathbb{E}_{\mu^\pi}\big[X_{\nu,\infty} \cdot [\![æ_{1:\infty} \in A]\!]\big].$$

This says that $X_{\nu,\infty}$ plays the role of the Radon–Nikodym derivative $\mathrm{d}\nu^\pi/\mathrm{d}\mu^\pi$ on the space of infinite histories. The proof rests on two ingredients beyond the scope of this worksheet:

- L^1 **martingale convergence** (closed martingale / Lévy's upward theorem): since $\mathbb{E}_{\mu^\pi}[X_{\nu,t}] = 1$ for every t, the non-negative μ^π-martingale $(X_{\nu,t})$ is uniformly integrable, so $X_{\nu,t} \to X_{\nu,\infty}$ in $L^1(\mu^\pi)$, not merely μ^π-a.s. This lets us pass the $t \to \infty$ limit through the expectation.
- **Uniqueness of measure extension** (Carathéodory / π–λ theorem): both sides of the identity are finite measures on $(\mathcal{A} \times \mathcal{E})^\infty$; the finite-

history identity (Exercise 10.1) shows they agree on all cylinder events $A = A_0 \times (\mathcal{A} \times \mathcal{E})^\infty$, and cylinders generate the full event σ-algebra, so agreement extends uniquely to every event.

We omit the proof and use this identity freely below.

Exercise 10.3 (Markov inequality for change of measure). **[15]** Let E be an event of infinite histories. For any $\varepsilon > 0$, show that

$$\mu^\pi\big[E \cap \{X_{\nu,\infty} \geq \varepsilon\}\big] \;\leq\; \frac{\nu^\pi[E]}{\varepsilon}.$$

Hint: On $E \cap \{X_{\nu,\infty} \geq \varepsilon\}$, $X_{\nu,\infty} \geq \varepsilon$ pointwise. Take μ^π-expectations and apply Exercise 10.2.

Solution to Exercise 10.3

Let $E_\varepsilon := E \cap \{X_{\nu,\infty} \geq \varepsilon\}$. On E_ε, $X_{\nu,\infty} \geq \varepsilon$, so pointwise

$$X_{\nu,\infty} \cdot [\![æ_{1:\infty} \in E_\varepsilon]\!] \;\geq\; \varepsilon \cdot [\![æ_{1:\infty} \in E_\varepsilon]\!].$$

Take μ^π-expectations and apply Exercise 10.2:

$$\nu^\pi[E_\varepsilon] \;=\; \mathbb{E}_{\mu^\pi}\big[X_{\nu,\infty} \cdot [\![æ_{1:\infty} \in E_\varepsilon]\!]\big] \;\geq\; \varepsilon \cdot \mathbb{E}_{\mu^\pi}\big[[\![æ_{1:\infty} \in E_\varepsilon]\!]\big] \;=\; \varepsilon \cdot \mu^\pi[E_\varepsilon].$$

Since $E_\varepsilon \subseteq E$, $\nu^\pi[E_\varepsilon] \leq \nu^\pi[E]$. Dividing by ε gives $\mu^\pi[E_\varepsilon] \leq \nu^\pi[E]/\varepsilon$.
Interpretation. A likelihood ratio of at least ε forces ν^π and μ^π measures of an event to be comparable: μ^π of the event can exceed ν^π of it only by the factor $1/\varepsilon$. In particular, if ν^π vanishes on E, so does μ^π on the part where the likelihood ratio stays bounded away from zero.

11. Proving the Self-Optimizing Property

Fix a policy $\tilde{\pi}$ that is self-optimizing for $\mathcal{M}$ with respect to π (Definition 9.1): its existence is the hypothesis of Theorem 9.2. For each $\nu \in \mathcal{M}$, define the **suboptimality gap**

$$\delta_{\nu,t} : \mathcal{H}^{t-1} \to [0,1], \qquad \delta_{\nu,t}(æ_{<t}) \;:=\; V_\nu^*(æ_{<t}) - V_\nu^{\tilde{\pi}}(æ_{<t}).$$

As with $X_{\nu,t}$, we write $\delta_{\nu,t}$ without its argument when convenient: "$\delta_{\nu,t} \to 0$ μ^π-a.s." means $\delta_{\nu,t}(æ_{<t}) \to 0$ on every infinite history outside a μ^π-null set, and similarly for "$\delta_{\nu,t} \leq 1$" etc.

Exercise 11.1 (Chain of inequalities). [20] Show:

$$0 \;\le\; w(\mu \mid æ_{<t})\big[V_\mu^* - V_\mu^{\pi_\xi^*}\big] \;\le\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\,\delta_{\nu,t}.$$

Hint: Use $V_\xi^{\pi_\xi^*} \ge V_\xi^{\tilde\pi}$ and Exercise 4.2.

Solution to Exercise 11.1

We establish two inequalities and chain them.
First inequality: isolate the μ-term. For each $\nu \in \mathcal{M}$, the optimal value is at least the value of any policy: $V_\nu^*(æ_{<t}) \ge V_\nu^{\pi_\xi^*}(æ_{<t})$, so $V_\nu^* - V_\nu^{\pi_\xi^*} \ge 0$. Since the posterior weights $w(\nu \mid æ_{<t}) \ge 0$, every term in $\sum_\nu w(\nu \mid æ_{<t})[V_\nu^* - V_\nu^{\pi_\xi^*}]$ is non-negative. The μ-term is one such term:

$$w(\mu \mid æ_{<t})\big[V_\mu^* - V_\mu^{\pi_\xi^*}\big] \;\le\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\big[V_\nu^* - V_\nu^{\pi_\xi^*}\big].$$

Second inequality: replace π_ξ^* with $\tilde\pi$. By definition of the optimal value, $V_\xi^*(æ_{<t}) \ge V_\xi^{\tilde\pi}(æ_{<t})$. Expanding both sides using infinite-horizon linearity (Exercise 4.2, applied to the fixed policies π_ξ^* and $\tilde\pi$):

$$\begin{aligned}\sum_\nu w(\nu \mid æ_{<t})\, V_\nu^{\pi_\xi^*}(æ_{<t}) \;&=\; V_\xi^*(æ_{<t}) \;\ge\; V_\xi^{\tilde\pi}(æ_{<t}) \\ &=\; \sum_\nu w(\nu \mid æ_{<t})\, V_\nu^{\tilde\pi}(æ_{<t}).\end{aligned}$$

Rearranging: $\sum_\nu w(\nu)[V_\nu^* - V_\nu^{\pi_\xi^*}] \le \sum_\nu w(\nu)[V_\nu^* - V_\nu^{\tilde\pi}] = \sum_\nu w(\nu)\,\delta_{\nu,t}$.
Chaining:

$$0 \;\le\; w(\mu \mid æ_{<t})\big[V_\mu^* - V_\mu^{\pi_\xi^*}\big] \;\le\; \sum_{\nu\in\mathcal{M}} w(\nu \mid æ_{<t})\,\delta_{\nu,t}.$$

Exercise 11.2 (Vanishing posterior). [10] Show: if $X_{\nu,\infty} = 0$ then $w(\nu \mid æ_{<t}) \to 0$ μ^π-a.s.

Hint: Show $w(\nu \mid æ_{<t}) = w_\nu X_{\nu,t-1}/X_{\xi,t-1}$ and use Exercise 9.3.

Solution to Exercise 11.2

From the posterior weight formula, dividing numerator and denominator by $\mu(æ_{<t})$:

$$w(\nu \mid æ_{<t}) \;=\; w_\nu \cdot \frac{\nu(æ_{<t})}{\xi(æ_{<t})} \;=\; w_\nu \cdot \frac{\nu(æ_{<t})/\mu(æ_{<t})}{\xi(æ_{<t})/\mu(æ_{<t})} \;=\; \frac{w_\nu \cdot X_{\nu,t-1}}{X_{\xi,t-1}},$$

where $X_{\nu,t-1}$ and $X_{\xi,t-1}$ are evaluated at the length-$(t-1)$ history $æ_{<t}$.
If $X_{\nu,\infty} = 0$, then $X_{\nu,t-1} \to 0$ μ^π-a.s. (Exercise 9.2). Since $X_{\xi,t-1} \geq w_\mu > 0$ always (Exercise 9.3), the ratio $w(\nu \mid æ_{<t}) = w_\nu X_{\nu,t-1}/X_{\xi,t-1} \to 0$ μ^π-a.s.

Exercise 11.3 (Transferring convergence). **[20]** Let $F := \{æ_{1:\infty} : \delta_{\nu,t}(æ_{<t}) \not\to 0\}$, the set of infinite histories on which the suboptimality gap fails to vanish. Show that $\mu^\pi\big[F \cap \{X_{\nu,\infty} > 0\}\big] = 0$.

Hint: Self-optimizing (Definition 9.1) gives $\nu^\pi[F] = 0$. Apply Exercise 10.3 with $E = F$ and $\varepsilon = 1/n$ to conclude $\mu^\pi[F \cap \{X_{\nu,\infty} \geq 1/n\}] = 0$ for each $n \geq 1$. Then note $\{X_{\nu,\infty} > 0\} = \bigcup_{n\geq 1}\{X_{\nu,\infty} \geq 1/n\}$ and take the countable union (Definition 7.1).

Solution to Exercise 11.3

Step 1: F is ν^π-null. By the self-optimizing assumption (Definition 9.1), under ν^π the suboptimality gap $\delta_{\nu,t}$ vanishes, so $\nu^\pi[F] = 0$.
Step 2: Apply Exercise 10.3. For each $n \geq 1$, take $E = F$ and $\varepsilon = 1/n$:

$$\mu^\pi\big[F \cap \{X_{\nu,\infty} \geq 1/n\}\big] \;\leq\; \frac{\nu^\pi[F]}{1/n} \;=\; 0.$$

Step 3: Countable union. The level sets of $X_{\nu,\infty}$ form a nested increasing family: $\{X_{\nu,\infty} > 0\} = \bigcup_{n\geq 1}\{X_{\nu,\infty} \geq 1/n\}$. Intersecting with F gives $F \cap \{X_{\nu,\infty} > 0\} = \bigcup_{n\geq 1}\big(F \cap \{X_{\nu,\infty} \geq 1/n\}\big)$, a countable union. By countable subadditivity (Definition 7.1):

$$\begin{aligned}\mu^\pi\big[F \cap \{X_{\nu,\infty} > 0\}\big] \;&= \mu^\pi\Big[\textstyle\bigcup_n F \cap \{X_{\nu,\infty} \geq 1/n\}\Big] \\ &\leq \sum_{n\geq 1} \mu^\pi\big[F \cap \{X_{\nu,\infty} \geq 1/n\}\big] \;=\; 0.\end{aligned}$$

What this says. "$\delta_{\nu,t} \to 0$ μ^π-a.s. on $\{X_{\nu,\infty} > 0\}$" is shorthand for exactly this event statement: among infinite histories on which the likelihood ratio stays bounded away from zero, all but a μ^π-null subset are ones where $\delta_{\nu,t}(æ_{<t}) \to 0$. We do *not* claim convergence on histories where $X_{\nu,\infty} = 0$: that case is handled separately in Exercise 11.2 via the posterior weight.

Exercise 11.4 (Single-term convergence). [10] Combine Exercise 11.2 and Exercise 11.3 to show $w(\nu \mid æ_{<t})\delta_{\nu,t} \to 0$ μ^π-a.s.

Solution to Exercise 11.4

Fix $\nu \in \mathcal{M}$. We show $w(\nu \mid æ_{<t})\,\delta_{\nu,t} \to 0$ μ^π-a.s. by considering two exhaustive cases:
Case 1: $X_{\nu,\infty} = 0$. By Exercise 11.2, $w(\nu \mid æ_{<t}) \to 0$. Since $\delta_{\nu,t} \in [0,1]$ (Exercise 1.3), the product $w(\nu \mid æ_{<t}) \cdot \delta_{\nu,t} \le 1 \cdot w(\nu \mid æ_{<t}) \to 0$.
Case 2: $X_{\nu,\infty} > 0$. By Exercise 11.3, $\delta_{\nu,t} \to 0$. Since $w(\nu \mid æ_{<t}) \le 1$ (posterior weights are at most 1), the product $w(\nu \mid æ_{<t}) \cdot \delta_{\nu,t} \le 1 \cdot \delta_{\nu,t} \to 0$.
These two cases cover all infinite histories (μ^π-a.s.), so $w(\nu \mid æ_{<t})\,\delta_{\nu,t} \to 0$ μ^π-a.s.

Exercise 11.5 (Self-Optimizing Theorem). [15] Combine Exercise 11.4, Exercise 11.1, Exercise 9.3 to finally prove the main result of Theorem 9.2.

Hint: $w(\mu \mid æ_{<t}) = w_\mu / X_{\xi,t-1}$.

Solution to Exercise 11.5

Since $\mathcal{M} = \{\nu_1, \dots, \nu_N\}$ is finite, we can sum Exercise 11.4 over all $\nu \in \mathcal{M}$:

$$\sum_{\nu \in \mathcal{M}} w(\nu \mid æ_{<t})\,\delta_{\nu,t} \;\longrightarrow\; 0 \qquad \mu^\pi\text{-a.s.}$$

From Exercise 11.1:

$$0 \;\le\; w(\mu \mid æ_{<t})\big[V^*_\mu(æ_{<t}) - V^{\pi^*_\xi}_\mu(æ_{<t})\big] \;\le\; \sum_\nu w(\nu \mid æ_{<t})\,\delta_{\nu,t} \;\longrightarrow\; 0.$$

So $w(\mu \mid æ_{<t})[V^*_\mu - V^{\pi^*_\xi}_\mu] \to 0$ μ^π-a.s.
It remains to divide by $w(\mu \mid æ_{<t})$. From Exercise 9.3:

$$w(\mu \mid æ_{<t}) \;=\; \frac{w_\mu}{X_{\xi,t-1}}.$$

Fix an infinite history outside the (single) μ^π-null set on which $X_{\xi,t} \to X_{\xi,\infty} < \infty$ fails. Along this history:

- $X_{\xi,t-1} \to X_{\xi,\infty}$, a finite positive number (with $X_{\xi,\infty} \ge w_\mu > 0$ by Exercise 9.3).
- Hence $w(\mu \mid æ_{<t}) = w_\mu / X_{\xi,t-1} \to w_\mu / X_{\xi,\infty} > 0$, so eventually $w(\mu \mid æ_{<t}) \ge w_\mu/(2X_{\xi,\infty}) > 0$.

Dividing the convergence $w(\mu \mid æ_{<t})(V_\mu^* - V_\mu^{\pi_\xi^*}) \to 0$ by this positive (per-history) lower bound, and using $V_\mu^* - V_\mu^{\pi_\xi^*} \geq 0$:

$$V_\mu^*(æ_{<t}) - V_\mu^{\pi_\xi^*}(æ_{<t}) \;\longrightarrow\; 0 \qquad \mu^\pi\text{-a.s.}$$

This completes the proof of Theorem 9.2.

Remark

We do not need to know which policy $\tilde{\pi}$ is self-optimizing, or whether it is computable. Mere existence suffices. For countable $\mathcal{M}$, this final step is harder: a countable sum of μ^π-a.s.-convergent sequences need not converge μ^π-a.s. The general proof uses a "convergence of mixture tails" argument [Hut05, Lem. 5.28].

Further reading

- Hutter, *An Introduction to Universal Artificial Intelligence* (2024): Chapter 2.7 (Kolmogorov complexity), Chapters 3.7–3.8 (the model class and universal prior), and Chapter 7.4 (AIXI).
- Hutter, *Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability* (Springer, 2005) — the original book-length treatment; Lem. 5.28 handles the countable-$\mathcal{M}$ self-optimizing case.
- Blackwell & Dubins, *Merging of Opinions with Increasing Information* (Ann. Math. Statist., 1962) — the merging-of-opinions theorem behind on-policy value convergence.
- Leike & Hutter, *Bad Universal Priors and Notions of Optimality* (COLT 2015) — adversarial choices of the universal Turing machine can make AIXI behave arbitrarily badly.

A. Worked Example: Bayesian Mixture and Value Function

Note

Setup.

- **Actions:** $\mathcal{A} = \{H, T\}$ (predict the next coin flip)
- **Observations:** $\mathcal{O} = \{H, T\}$ (actual coin flip)
- **Rewards:** $\mathcal{R} = \{0, 1\}$, with $r_t = [\![a_t = o_t]\!]$
- **Model class:** $\mathcal{M} = \{\nu_{HH}, \nu_{HT}\}$ (two-headed coin, fair coin)
- **Prior:** $w_{\nu_{HH}} = w_{\nu_{HT}} = \frac{1}{2}$

Before any interaction ($t = 1$, $æ_{<1} = \epsilon$):

$$\xi(o_1 = H \mid a_1) \;=\; \tfrac{1}{2} \cdot 1 + \tfrac{1}{2} \cdot \tfrac{1}{2} \;=\; \tfrac{3}{4}.$$

After observing a head ($t = 2$), the posterior updates:

$$w(\nu_{HH} \mid æ_1) \;=\; \tfrac{1}{2} \cdot \frac{1}{3/4} \;=\; \tfrac{2}{3}, \qquad w(\nu_{HT} \mid æ_1) \;=\; \tfrac{1}{2} \cdot \frac{1/2}{3/4} \;=\; \tfrac{1}{3}.$$

Updated prediction: $\xi(o_2 = H \mid æ_1 a_2) = \frac{2}{3} \cdot 1 + \frac{1}{3} \cdot \frac{1}{2} = \frac{5}{6}$.

Value function. Continuing the same setup, suppose the agent always predicts H (policy π_H).

Under ν_{HH}: always correct, $r_t = 1$ every step: $V^{\pi_H}_{\nu_{HH}}(\epsilon) = (1-\gamma)\sum_{k=0}^{\infty} \gamma^k \cdot 1 = 1$.

Under ν_{HT}: correct half the time: $V^{\pi_H}_{\nu_{HT}}(\epsilon) = (1-\gamma)\sum_{k=0}^{\infty} \gamma^k \cdot \frac{1}{2} = \frac{1}{2}$.

The mixture value (by Exercise 4.2) is: $V^{\pi_H}_{\xi}(\epsilon) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}$.

As the agent observes more heads ($\mu = \nu_{HH}$), the posterior on ν_{HH} increases towards 1, and $V^{\pi_H}_{\xi} \to V^{\pi_H}_{\nu_{HH}} = 1$. This is on-policy value convergence (Theorem 7.6) in action.

B. Knuth's Difficulty Scale

Each subproblem carries a difficulty rating in square brackets, following Knuth's rating scheme for exercises [Knu73] in slightly adapted form. The rating assumes that the material in the preceding problems (on which the subproblem depends) has been understood. In-between values are possible.

[00] *Very easy.* Solvable from the top of your head.

[10] *Easy.* Needs 15 minutes to think, possibly pencil and paper.

[20] *Average.* May take 1–2 hours to answer completely.

[30] *Moderately difficult or lengthy.* May take several hours to a day.

[40] *Quite difficult or lengthy.* Often a significant research result.

[50] *Open research problem.* An obtained solution should be published.

References

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